MODE STABILITY FOR GRAVITATIONAL PERTURBATIONS OF SUBEXTREMAL KERR–DE SITTER BLACK HOLES

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Abstract. A Kerr–de Sitter black hole is mode stable if no linear perturbation grows exponentially or oscillates without decay. Hintz proved this for scalar waves in the full subextremal range. We prove it for the Teukolsky equations of spin ± 2, and deduce that every generalized mode of the ungauged linearized Einstein equation with $\operatorname{Im}\sigma \geq 0$ is pure gauge, up to a linearized Kerr–de Sitter metric when $\sigma = 0$. This is the hypothesis under which Hintz, Petersen and Vasy proved the nonlinear stability of Kerr–de Sitter in the full subextremal range. Part of the proof is computer-assisted.

1. Introduction

For $\Lambda > 0$, $M > 0$ and $a \in \mathbb{R}$, the Kerr–de Sitter (KdS) metric in Boyer–Lindquist coordinates is

$$g = -\frac{\Delta_r}{\Xi^2 \varrho^2}\big(dt - a\sin^2\theta\, d\varphi\big)^2 + \varrho^2\Big(\frac{dr^2}{\Delta_r} + \frac{d\theta^2}{\Delta_\theta}\Big) + \frac{\Delta_\theta \sin^2\theta}{\Xi^2\varrho^2}\big((r^2+a^2)\, d\varphi - a\, dt\big)^2, \tag{1.1}$$

where

$$\Delta_r = (r^2+a^2)\Big(1-\frac{\Lambda r^2}{3}\Big) - 2Mr, \quad \varrho^2 = r^2 + a^2\cos^2\theta, \quad \Delta_\theta = 1 + \alpha\cos^2\theta, \quad \alpha = \frac{\Lambda a^2}{3}, \quad \Xi = 1+\alpha.$$

It solves $\operatorname{Ric}(g) = \Lambda g$. The parameters are *subextremal* if Δ_r has four distinct real roots, which we label

$$r_3 < 0 \leq r_0 < r_1 < r_2,$$

where $r_0 = r_C$ is the Cauchy horizon, $r_1 = r_e$ the event horizon and $r_2 = r_c$ the cosmological horizon; then $r_0 + r_1 + r_2 + r_3 = 0$. The domain of outer communications is $r_1 < r < r_2$. We use a time function t_* and an azimuthal angle φ_* that are smooth across the future event and cosmological horizons, as in [4, 3]; a *mode* at frequency $\sigma \in \mathbb{C}$ is a function, or a tensor, of the form $e^{-i\sigma t_*}v$ with v independent of t_* and smooth on $[r_1 - \delta, r_2 + \delta] \times S^2$. In the language of [3], such modes are *outgoing*.

Hintz and Vasy proved in 2018 that the KdS family is nonlinearly stable for small a/M [5]. Hintz, Petersen and Vasy extended this to the full subextremal range, conditional on mode stability of the ungauged linearized Einstein equation [4, Assumption 1.2]. Hintz then proved mode stability for scalar waves in the same range [3], and left spin ± 2 open. We settle that case.

Theorem 1.1 (Teukolsky mode stability)**.** *On every subextremal KdS spacetime, the Teukolsky equations of spin $s = +2$ and $s = -2$ have no nontrivial outgoing mode solutions at any frequency $\sigma \in \mathbb{C}$ with $\operatorname{Im}\sigma \geq 0$, including $\sigma = 0$. The same holds for generalized modes $\sum_{j<k} t_*^j e^{-i\sigma t_*} v_j$.*

Theorem 1.2 (Ungauged linearized Einstein mode stability)**.** *Let (Λ, M, a) be subextremal with $a \neq 0$, and let $T = \partial_{t_*} + \frac{a}{r_2^2+a^2}\partial_{\varphi_*}$. Let h be a smooth symmetric 2-tensor with $D_g\operatorname{Ric}(h) = \Lambda h$.*

(1) If $(\mathcal{L}_T + i\sigma)^k h = 0$ for some $k \geq 1$, with $\sigma \neq 0$ and $\operatorname{Im}\sigma \geq 0$, then $h = \mathcal{L}_X g$ for a smooth vector field X.

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(2) *If* $(\mathcal{L}_T)^k h = 0$ *for some* $k \geq 1$*, then* $h = c_M \partial_M g + c_a \partial_a g + \mathcal{L}_X g$ *with* $c_M, c_a \in \mathbb{C}$ *and* X *smooth.*

For $a = 0$, Theorem 1.2 is [5, §7]. Theorem 1.2 is [4, Assumption 1.2] (see Remark 1.4), so [4, Theorem 1.4] gives:

Corollary 1.3. *Subextremal Kerr–de Sitter black holes are nonlinearly stable as solutions of the Einstein vacuum equations with positive cosmological constant, in the sense of* [4, Theorem 1.4].

Remark 1.4 (Choice of T). [4] use $T' = T + c\,\partial_{\varphi_*}$ for some $c \in \mathbb{R}$. Since ∂_{φ_*} commutes with T, T' and $D_g\mathrm{Ric} - \Lambda$, a generalized T'-mode h at σ' can be put in the gauge of [4, (23)] [5, §10] and then lies in a finite-dimensional, ∂_{φ_*}-invariant space of resonant states. So $h = \sum_{|m| \leq m_0} h_m$ with each h_m a generalized T-mode at $\sigma' - cm$, which has the same imaginary part. Theorem 1.2 applies to each h_m (for generalized modes with $\sigma \neq 0$, via the reduction in [4, Theorem 2.7]); if $\sigma' = cm$ with $m \neq 0$, the linearized KdS part vanishes by axisymmetry.

1.1. **Outline.** For real $\sigma \neq 0$ we separate variables. Outside the superradiant window a Wronskian identity is enough. Inside it we do not work with the radial Teukolsky equation directly. An Euler transform of its Heun form gives a dual equation which, unlike the original, is real, with potential $L(z) = A/z + B - 3z$ and $A > 0$; the horizons sit at $z = 1$ and $z = t$. This is the structure of the scalar problem in [3], and Hintz's two currents reduce everything to an ordinary differential inequality (the *event lemma*, §4) and a condition at the cosmological horizon (Proposition 3.10). Both come down to signs of explicit polynomials. At $\sigma = 0$ an energy identity and a Teukolsky–Starobinsky map suffice (§5).

For $\mathrm{Im}\,\sigma > 0$ we deform to Schwarzschild–de Sitter, as Hintz does for scalars (§6). The difficulty is that no high-frequency estimate for Teukolsky is known in the full subextremal range. We borrow one. A Teukolsky mode, used as a Hertz potential [13], gives a linearized Einstein mode; if its Ψ_4 is nonzero, it becomes a quasinormal mode of the gauge-fixed operator of [4], whose high-energy estimates confine it to a compact set. That $\Psi_4 \neq 0$ holds off a discrete set of frequencies, because nothing killed by $(\text{þ}')^4$ can be regular at the cosmological horizon.

Theorem 1.2 follows from Theorem 1.1 in §7. Once the Teukolsky scalars vanish, the perturbation is algebraically special to first order, and the Killing spinor field ξ and Carter's field ζ stay Killing. This makes h pure gauge with an explicit gauge vector, except in the stationary axisymmetric sector, where Carter's separable ansatz linearizes the field equations.

Computer assistance. The computer-assisted steps are listed in Appendix A. All are exact except C9, which uses floating-point arithmetic with rigorous error bounds. The code is available on request.

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2. Preliminaries

2.1. **Parameters.** By Vieta's formulas,

$$\sum_j r_j = 0, \qquad \sum_{i<j} r_i r_j = -\frac{3(1-\alpha)}{\Lambda}, \qquad r_0 r_1 r_2 r_3 = -\frac{3a^2}{\Lambda}. \tag{2.1}$$

We write $\rho_j = r_j^2 + a^2$, $\Omega_j = a/\rho_j$, $D_j = |\Delta_r'(r_j)|$ and $P_j = \rho_j / D_j$. The signs of $\Delta_r'(r_j)$ are $(-,+,-,+)$ for $j = 0,1,2,3$. Following [3, §6, footnote 5], we use the roots as parameters.

Lemma 2.1 (Rotation bound)**.** *On every subextremal KdS spacetime,* $\alpha < 7 - 4\sqrt{3}$.

Proof. By scaling we may assume $\Lambda = 3$. Then $-\Delta_r = r^4 - (1-\alpha)r^2 + 2Mr - \alpha$, so $p_2 := \sum r_j^2 = 2(1-\alpha)$ and $\mathcal{A} := -r_0r_1r_2r_3 = \alpha$. Scaling a configuration of roots by $\tau > 0$ multiplies $\mathcal{A}$ by τ^4 and p_2 by τ^2. Hence α is the positive root of $\alpha = 1 - \frac{1}{2}\sqrt{\alpha/k}$, where $k := \mathcal{A}/p_2^2$ is scale invariant, and this root is increasing in k. Put $S = -r_3 = r_0 + r_1 + r_2$. By the AM–GM inequality, $r_0r_1r_2 \le S^3/27$, and by the QM–AM inequality, $r_0^2 + r_1^2 + r_2^2 \ge S^2/3$. Thus $k \le (S^4/27)/(16S^4/9) = 1/48$, with equality only if $r_0 = r_1 = r_2$, which subextremality excludes. At $k = 1/48$ the root is $\alpha = 12(3^{-1/2} - \frac{1}{2})^2 = 7 - 4\sqrt{3}$. □

2.2. **The Teukolsky equations and separation of variables.** For σ real, the Teukolsky equation of spin s separates as $e^{-i\sigma t + im\varphi}R(r)S(\theta)$ [8]. The angular operator is then symmetric with compact resolvent. As in [3, §3], a mode solution can therefore be decomposed into separated ones, each with an eigenvalue $\lambda \in \mathbb{R}$. We only separate variables for real σ. Put

$$K(r) = \Xi\big((r^2+a^2)\sigma - am\big), \qquad \kappa_j = \frac{iK(r_j)}{\Delta_r'(r_j)}.$$

The residue theorem applied to ρ/Δ_r, $1/\Delta_r$ and r^k/Δ_r gives

$$\sum_j \kappa_j = 0, \qquad \sum_j r_j\kappa_j = -\frac{3i\Xi\sigma}{\Lambda}, \qquad \sum_j r_j^2\kappa_j = 0, \tag{2.2}$$

together with $P_1 + P_3 = P_0 + P_2$ and $D_1^{-1} + D_3^{-1} = D_0^{-1} + D_2^{-1}$.

Lemma 2.2 (Boundary conditions)**.** *A separated mode is outgoing if and only if*

$$R \in (r-r_1)^{-\kappa_1 - s}\, C^\infty([r_1, r_2)) \cap (r_2 - r)^{\kappa_2}\, C^\infty((r_1, r_2]).$$

Proof. Write $t = t_* + T(r)$ and $\varphi = \varphi_* + \Phi(r)$ as in [3, (3.4)]. Near r_j ($j = 1, 2$) the singular parts are $T' = \mp\Xi\rho_j/\Delta_r + O(1)$ and $\Phi' = \mp\Xi a/\Delta_r + O(1)$, with the sign chosen so that t_*, φ_* are regular across the future horizon (upper sign at r_1, lower at r_2). A mode is $e^{-i\sigma t_* + im\varphi_*}v = e^{-i\sigma t + im\varphi}\, e^{i\sigma T - im\Phi}v$, so its radial part is $R = e^{i\sigma T - im\Phi}v$ (after separating the angular part), and $i\sigma T - im\Phi = \mp\frac{i\,\Xi(\rho_j\sigma - am)}{\Delta_r'(r_j)}\log|r - r_j| + (\text{smooth}) = \mp\kappa_j \log|r - r_j| + (\text{smooth})$. Hence v smooth gives $R \in |r-r_1|^{-\kappa_1}C^\infty$ at r_1 and $R \in |r-r_2|^{\kappa_2}C^\infty$ at r_2, for $s = 0$. The additional factor Δ_r^{-s} at r_1 comes from the boost $l = \Delta_r l_K$, $n = \Delta_r^{-1}n_K$ of Lemma 6.1, which is regular across the event horizon: the Teukolsky scalar of spin s has boost weight s, so its Kinnersley component is Δ_r^{-s} times the regular one. The Kinnersley tetrad is already regular across the future cosmological horizon in outgoing coordinates. □

2.3. **Heun form.** The Möbius map z with $z(r_0, r_1, r_2, r_3) = (0, 1, t, \infty)$, where

$$t = \frac{(r_1 - r_3)(r_2 - r_0)}{(r_1 - r_0)(r_2 - r_3)} > 1,$$

turns the radial Teukolsky equation into a Heun equation with singular points $0, 1, t, \infty$. We write $\mu(z) = z(z-1)(t-z)$. The Heun-form function is $y = \prod_{j\in\{0,1,t\}}(z - z_j)^{-\kappa_j}R$ (up to a constant factor). By Lemma 2.2 an outgoing mode has $y \in (z-1)^{-2\kappa_1 - s}C^\infty$ near $z = 1$ and y smooth at $z = t$; the exponents at ∞ are $\{1 + 2s,\, 1 + s - 2\kappa_3\}$. Two constants will appear repeatedly:

$$C_0 = \frac{3}{\Lambda(r_1 - r_0)(r_2 - r_3)}, \qquad c_1 = 1 + t + 3(1-\alpha)C_0. \tag{2.3}$$

3. Real frequencies: the radial problem

Throughout this section $\sigma \in \mathbb{R} \setminus \{0\}$. We fix a separated outgoing mode with eigenvalue λ. By the symmetries $(\sigma, m, a) \mapsto (-\sigma, -m, a)$ and $(m, a) \mapsto (-m, -a)$ (reflection $\varphi \mapsto -\varphi$) we may assume $m \geq 0$ and $a \geq 0$. For $a = 0$ the window below is empty ($w_1 = w_2 = 0$), so in the window $a > 0$.

3.1. **The superradiant window.** Set $e_1 = 2(\kappa_1 + \kappa_3)$ and $e_t = 2(\kappa_2 + \kappa_3)$. For real σ the κ_j are purely imaginary.

Proposition 3.1 (The window)**.** *Let $m > 0$ and $w = \sigma/m$. Then $\operatorname{Im} e_1 = 2\Xi m f_1(w)$ and $\operatorname{Im} e_t = 2\Xi m f_2(w)$, where*

$$f_1(w) = P_1(w - \Omega_1) + P_3(w - \Omega_3), \qquad f_2(w) = -P_2(w - \Omega_2) + P_3(w - \Omega_3).$$

The function f_1 is increasing and f_2 is decreasing, and their zeros satisfy $0 < w_2 < w_1$. Consequently $\operatorname{Im} e_1 \cdot \operatorname{Im} e_t > 0$ exactly on the open interval (w_2, w_1), the window*.*

Proof. The formulas follow from the definition of κ_j and the signs of $\Delta_r'(r_j)$. The slope of f_1 is $P_1 + P_3 > 0$, and by the residue identities that of f_2 is $P_3 - P_2 = P_0 - P_1$. Now $D_0/D_1 = f(r_0)/f(r_1)$ for the concave parabola $f(x) = (x - r_3)(r_2 - x)$, whose vertex $-(r_0 + r_1)/2$ lies left of r_0; so $D_0 > D_1$, and as $\rho_1 > \rho_0$, $P_1 > P_0$. The same comparison gives $w_2 = a(D_1^{-1} - D_0^{-1})/(P_1 - P_0) > 0$. Finally, a direct computation gives

$$w_1 - w_2 = \frac{2a(r_2 - r_1)(r_0 - r_3)}{(2a^2 - r_2 r_3 - r_0 r_1)(2a^2 - r_1 r_3 - r_0 r_2)},$$

and every factor is positive, since $-r_2 r_3 = r_2(r_0 + r_1 + r_2) > r_0 r_1$ and $-r_1 r_3 = r_1(r_0 + r_1 + r_2) > r_0 r_2$. □

For every real $\sigma \neq 0$, Proposition 3.5 below puts the dual equation in the real form (3.2), and u is outgoing at both ends. Let B_1, B_t be the boundary coefficients of u at $z = 1, t$ in the sense of [3, (3.5)]. Conservation of $\mu \operatorname{Im}(u' \bar{u})$ gives, as in [3, §3],

$$\operatorname{Im} e_1 \, |B_1|^2 = t \, \operatorname{Im} e_t \, |B_t|^2. \tag{3.1}$$

Corollary 3.2. *There are no outgoing modes with $\sigma \in \mathbb{R} \setminus \{0\}$ outside the open window, nor any with $m = 0$.*

Proof. For $m = 0$, $\operatorname{Im} e_1 = 2\Xi\sigma(P_1 + P_3)$ and $\operatorname{Im} e_t = 2\Xi\sigma(P_3 - P_2) = -2\Xi\sigma(P_1 - P_0)$ have opposite signs. Off the closed window the same holds by Proposition 3.1, so (3.1) forces $B_1 = B_t = 0$. At the endpoints exactly one of $\operatorname{Im} e_1, \operatorname{Im} e_t$ vanishes, since $w_1 \neq w_2$, and (3.1) gives $B_t = 0$ or $B_1 = 0$ at the other end. In each case the indicial roots at that end differ by the non-integer e_t or e_1, so $u = 0$ as after [3, (3.6)]. Then $\tilde{y} = 0$, which contradicts Proposition 3.4. The cases excluded there, $s = +2$ with $\sigma \in \{m\Omega_1, m\Omega_3\}$, follow from the case $s = -2$ by Lemma 3.3, since $\Omega_1, \Omega_3 \neq \Omega_2$. □

3.2. **The dual equation.** Two exceptional frequencies of spin $+2$ are first reduced to spin -2.

Lemma 3.3 (Reduction of spin $+2$ to spin -2)**.** *Let $\sigma \in \mathbb{R}$ with $\kappa_2 \neq 0$, i.e. $\sigma \neq m\Omega_2$. If outgoing $s = -2$ modes at (σ, m, λ) vanish, so do outgoing $s = +2$ modes.*

Proof. Let $D^\dagger = \partial_r + iK/\Delta_r$ and $\psi = \Delta_r^2 (D^\dagger)^4 (\Delta_r^2 R)$. This maps solutions of the $s = +2$ radial equation to solutions of the $s = -2$ radial equation with the same λ (an exact identity, item C10; the spin ± 2 angular eigenvalues agree because the angular operator is invariant under $(s, \cos\theta) \mapsto (-s, -\cos\theta)$). Near r_1, $iK/\Delta_r = \kappa_1/(r - r_1) + O(1)$, so $D^\dagger$ maps $(r - r_1)^{-\kappa_1} C^\infty$ to itself; since $\Delta_r^2 R \in (r - r_1)^{-\kappa_1} C^\infty$ by Lemma 2.2, $\psi \in (r - r_1)^{2 - \kappa_1} C^\infty$, the $s = -2$ outgoing

condition. Near r_2, $iK/\Delta_r = -\kappa_2/(r_2-r)+O(1)$, so $D^\dagger$ maps $(r_2-r)^\beta C^\infty$ into $(r_2-r)^{\beta-1}C^\infty$, and $\psi \in (r_2-r)^{\kappa_2}C^\infty$ is outgoing. Hence $\psi = 0$, i.e. $\Delta_r^2 R = e^{-i\int K/\Delta_r} p$ with p a cubic polynomial. Near r_2 this lies in $(r_2-r)^{-\kappa_2}C^\omega$, which meets $(r_2-r)^{\kappa_2+2}C^\infty$ only in 0 unless $2\kappa_2 \in \mathbb{Z}$. But $\kappa_2 \in i\mathbb{R}\setminus\{0\}$. So $R = 0$. □

Proposition 3.4 (Euler transform)**.** *Let $\sigma \in \mathbb{R}$ with $\sigma \neq 0$ or $am \neq 0$, and if $s = +2$ assume $\kappa_1\kappa_3 \neq 0$. Let $y \neq 0$ be the Heun-form radial function of an outgoing mode, normalized so that it is analytic at $z = t$ and lies in $(z-1)^{\rho_1}C^\omega$ near $z = 1$, where $\rho_1 = -2\kappa_1 - s$. Put $\nu = s-1-2\kappa_3$. Then $\tilde y(z) = \int_\gamma (z-w)^\nu y(w)\,dw$, for a suitable contour γ, is a nonzero solution of a Heun equation $\widetilde{\mathcal{H}}\tilde y = 0$ such that:*

(i) the exponents of $\widetilde{\mathcal{H}}$ at $z_j \in \{0,1,t\}$ are 0 and $-2(\kappa_j+\kappa_3)$;
(ii) $\tilde y$ is analytic at $z = t$, and $\tilde y \in (z-1)^{-e_1}C^\omega$ near $z = 1$, where $e_1 = 2(\kappa_1+\kappa_3)$.

Proof. Write $\mathcal{H} = P_2\partial_w^2 + P_1\partial_w + P_0$ with $P_2 = w(w-1)(w-t)$. Since $\nu+2$ is an exponent at ∞, there is a Heun operator $\widetilde{\mathcal{H}}$ with $\widetilde{\mathcal{H}}_z(z-w)^\nu = \mathcal{H}_w^\dagger(z-w)^\nu$ [6, 12], and (i) follows from the exponents at ∞. With $K = (z-w)^\nu$,

$$K\,\mathcal{H}y - y\,\mathcal{H}_w^\dagger K = \partial_w\mathcal{B}, \qquad \mathcal{B} = P_2(Ky' - K_w y) + (P_1 - P_2')Ky,$$

so $\widetilde{\mathcal{H}}\tilde y = 0$ whenever the boundary contribution of $\mathcal{B}$ along γ vanishes. We take γ inside a neighborhood of $[1,z]$ on which y is analytic (z near t); then $\tilde y$ is analytic in z, in particular at t. Here $\nu \notin \mathbb{Z}_{\geq 0}$: for $s = +2$, $\nu = 1-2\kappa_3$ with $\kappa_3 \in i\mathbb{R}\setminus\{0\}$, and for $s = -2$, $\operatorname{Re}\nu = -3$. For $\beta = \rho_1 + k$ we have $\beta+\nu+2 = k+1-e_1$ with $e_1 \in i\mathbb{R}$.

$\kappa_1 \neq 0$. Take the Pochhammer loop $\gamma = [1^+, z^+, 1^-, z^-]$; it is closed, so $\mathcal{B}$ contributes nothing. On powers, $\oint_\gamma (z-w)^\nu(w-1)^\beta dw = c(\beta)(z-1)^{\beta+\nu+1}$ with $c(\beta) = \pm(2\pi i)^2 e^{i\pi(\beta+\nu)}/\big(\Gamma(-\beta)\Gamma(-\nu)\Gamma(\beta+\nu+2)\big)$, which vanishes only if $\beta \in \mathbb{Z}_{\geq 0}$, $\nu \in \mathbb{Z}_{\geq 0}$ or $\beta+\nu+2 \in \mathbb{Z}_{\leq 0}$. Since $\rho_1 \notin \mathbb{R}$, none occurs for $\beta = \rho_1 + k$, and termwise integration gives $\tilde y = (z-1)^{-e_1}\sum_k c(\rho_1+k)a_k(z-1)^k \neq 0$.

$\kappa_1 = 0$, $s = -2$. Then $\kappa_3 \neq 0$ (as $\Omega_1 \neq \Omega_3$), so $\nu \notin \mathbb{Z}$. Now $\rho_1 = 2$ and $y = O((w-1)^2)$, so $\mathcal{B}(1) = 0$ (as $P_2(1) = 0$). Take γ a loop based at $w = 1$ encircling $w = z$ once; the integrand returns multiplied by $e^{2\pi i\nu}$, so the boundary contribution is $(e^{2\pi i\nu}-1)\mathcal{B}(1) = 0$. On powers the loop gives a nonzero multiple of $(1-e^{2\pi i\nu})\Gamma(\beta+1)\Gamma(\nu+1)/\Gamma(\beta+\nu+2)$, nonzero for $\beta = 2+k$; termwise integration again gives $\tilde y \in (z-1)^{-e_1}C^\omega\setminus\{0\}$. □

Proposition 3.5 (The real dual)**.** *For real σ with $\sigma \neq 0$ or $am \neq 0$, after multiplication by an explicit nonvanishing gauge factor, the equation $\widetilde{\mathcal{H}}\tilde y = 0$ becomes*

$$-(\mu u')' - \frac{q(z)^2}{\mu}u + L(z)u = 0, \qquad q(z) = q_1(z-z_0), \qquad L(z) = \frac{A}{z} + B - 3z, \tag{3.2}$$

with real coefficients, q affine, and $q(1)$, $q(t)$ equal, up to a common sign, to $\frac{1}{2}(t-1)\operatorname{Im}e_1$ and $-\frac{1}{2}t(t-1)\operatorname{Im}e_t$. If w lies in the open window, then moreover: $q = q_1(z-z_0)$ with $q_1 > 0$ and $z_0 \in (1,t)$, after replacing u by $\bar u$ and q by $-q$ if necessary;

$A = t\big(X_0^2 - (X_1+X_t)^2\big)$ with $X_j = |\kappa_j+\kappa_3|$, and $A > 0$;

$B = \lambda C_0 + c_1$. The transformed solution u is outgoing for (3.2) in the sense of [3, (3.4)]: near $z = 1$ it is $(z-1)^{-iq(1)/\mu'(1)}$ times a smooth function, and near $z = t$ it is $(t-z)^{-iq(t)/|\mu'(t)|}$ times a smooth function.

Proof. For real σ all exponents of $\widetilde{\mathcal{H}}$ at $0,1,t$ are purely imaginary, and the gauge factor $\prod_j(z-z_j)^{-\operatorname{Re}(\cdot)}$ removes the real parts. The resulting operator is formally real, and writing it in the form (3.2) determines $q(1)$ and $q(t)$ from the exponents at $z = 1$ and $z = t$. We take q to be linear. By Proposition 3.1, $q(1)q(t) < 0$, so q has a unique zero $z_0 \in (1,t)$, and the sign of q_1 is fixed by replacing u with $\bar u$ if necessary. The remaining claims are as follows. *Form of L.*

Writing the gauged operator as $-(\mu u')' - \mu V u$, the function $L := (q^2 - \mu^2 V)/\mu$ is rational, and $L = n_0/z + n_1 + n_2 z$. Expanding at $z = \infty$ with the exponents $\{1 + 2s, 1 + s - 2\kappa_3\}$ gives $n_2 = -3$, since the exponent difference there is ± 4.

The coefficient n_0. Expanding at $z = 0$ gives $n_0 = t(X_0^2 - (X_1 + X_t)^2)$. In the window $X_0 - X_1 - X_t = 2 \operatorname{Im} \kappa_3$, so $n_0 = 2t \operatorname{Im} \kappa_3 (X_0 + X_1 + X_t)$. This is positive because $\operatorname{Im} \kappa_3 > 0$ for $w > \Omega_3$, and $\Omega_3 < \Omega_2 < w$ since $|r_3| > r_2$.

The coefficient n_1. After eliminating by (2.1) and (2.2), $n_1 = \lambda C_0 + c_1$ is an identity of rational functions; the numerators of $n_1 - (\lambda C_0 + c_1)$ and $n_2 + 3$ are divisible by $s^2 - 4$ (item C1).

Outgoing behavior. This is Proposition 3.4(ii), after matching exponents. □

Lemma 3.6 (A priori bounds)**.** *Throughout the open window, $L' < 0$ on $[1, t]$, and for every $m \geq 1$*

$$L(z_0) \;\geq\; \frac{A}{2z_0} + 2t. \tag{3.3}$$

Proof. First, $L' = -A/z^2 - 3 < 0$. For (3.3) we use only a crude lower bound for the angular eigenvalue, $\lambda \geq \lambda_{\mathrm{cr}} := \ell_0(\ell_0{+}1){-}4{-}4\alpha{-}4\alpha|m|(1{+}\alpha){-}2(1{+}\alpha)^2|\xi|(|m|{+}2)$ with $\ell_0 = \max(|m|, 2)$ (Proposition 3.7). Since $L(z_0)$ is increasing in $B = \lambda C_0 + c_1$, it suffices to prove (3.3) with $\lambda = \lambda_{\mathrm{cr}}$. Then $\Lambda/3$ times $L(z_0) - A/(2z_0) - 2t$ is a quadratic polynomial $m^2 Q_2 + m Q_1 + Q_0$ (with the modification $+4C_0\Lambda/3$ at $m = 1$) whose coefficients are rational functions of (r_0, r_2, a^2, τ), where $\tau \in (0, 1)$ parametrizes the window. One has $Q_0 < 0$, so it suffices to show positivity at $m = 1$, at $m = 2$ and of Q_2. After eliminating a^2 by the quadratic relation it satisfies, each of these reduces to the positivity of finitely many polynomials on the open cube in $(r_0, 1/r_2, \tau)$. All of them have nonnegative Bernstein coefficients on $[0, 1]^3$, not all zero, which proves positivity on the open cube (Appendix A, item C2). □

Proposition 3.7 (Crude angular bound)**.** *For $s = \pm 2$, $m \geq 0$, $\alpha \geq 0$ and $\xi = a\sigma \geq 0$ (in particular on the window), $\lambda \geq \lambda_{\mathrm{cr}}$.*

Proof. In the variable $x = \cos\theta$, the angular operator of [11, (2.3)] is $H = H_0 + K_\alpha + V$, where

$$H_0 = -\partial_x(1 - x^2)\partial_x + \frac{(m + sx)^2}{1 - x^2}, \qquad K_\alpha = -\partial_x\, \alpha x^2(1 - x^2)\, \partial_x \;\geq 0,$$

and λ is the lowest eigenvalue of H in the sector (m, s). On that sector $H_0 \geq \ell_0(\ell_0 + 1) - s^2$ with $\ell_0 = \max(|m|, |s|)$. An exact computation gives $V = \frac{\alpha(m+s)^2}{2(1-x)} + \frac{\alpha(m-s)^2}{2(1+x)} + \frac{\mathcal{N}}{1+\alpha x^2}$, where the first two terms are nonnegative and

$$\mathcal{N} = \alpha^2 m^2 {-} 2\alpha^2 msx {+} 2\alpha^2 x^4 {+} \alpha m^2 {-} 2\alpha msx {+} \alpha s^2 x^2 {-} \alpha s^2 {+} 2\alpha x^2 {+} \xi^2(1{+}\alpha)^2(1{-}x^2) {-} 2\xi(1{+}\alpha)^2(m{-}sx).$$

Let $T = -4\alpha - 4\alpha m(1 + \alpha) - 2(1 + \alpha)^2\xi(m + 2)$. With $s^2 = 4$,

$$\begin{aligned}\mathcal{N} - T(1 + \alpha x^2) = {} & (2\alpha^2 x^4 + 4\alpha^2 x^2 + 6\alpha x^2) + (\alpha^2 + \alpha)m^2 \\ & + m\big(4\alpha^3 x^2 + 4\alpha^2 x^2 + 4(\alpha^2 + \alpha)(1 - \tfrac{s}{2}x)\big) + \xi^2(1 + \alpha)^2(1 - x^2) \\ & + \xi\big(2\alpha^3 m x^2 + 4\alpha^3 x^2 + 4\alpha^2 m x^2 + 8\alpha^2 x^2 + 2\alpha m x^2 + 4\alpha x^2 + 2(1 + \alpha)^2(2 + sx)\big).\end{aligned}$$

Every term is nonnegative for $|x| \leq 1$, $|s| = 2$, $m, \alpha, \xi \geq 0$. Hence $V \geq T$, and $\lambda \geq \ell_0(\ell_0 + 1) - 4 + T = \lambda_{\mathrm{cr}}$. □

3.3. **Currents.** On the dual equation (3.2) we use the currents of [3, Lemma 4.1], in the form with variable potential from [3, §7]. Let $u_\pm = \exp(\mp i \int_{z_0}^{z} q/\mu)\, u$; these are smooth down to $z = t$

and $z = 1$, respectively. For a real function f, define

$$J^f[v] = -\mu f q|v'|^2 - \mu f q_1 \operatorname{Re}(v'\bar{v}) + \big(Lfq + \tfrac{1}{2}q_1\mu f'\big)|v|^2,$$
$$C^f = \mu' f q - 2\mu f q_1 - \mu f' q, \qquad D^f = q(fL)' + \tfrac{1}{2}q_1(\mu f')'.$$

Then $\frac{d}{dz}J^f[u_\pm] = C^f|u_\pm'|^2 + D^f|u_\pm|^2$. The proof is the integration by parts of [3, Lemma 4.1], using $q' = q_1$.

Lemma 3.8 (Horizon inequality)**.** *For all $z_0 \in [1,t]$ and $z \in (1,t)$, we have $H(z) := 2\mu(z) + (z_0 - z)\mu'(z) > 0$.*

Proof. H is affine in z_0. At $z_0 = 1$ it equals $(z-1)\big((z-1)^2 + t - 1\big) > 0$. At $z_0 = t$ it equals $(t-z)\big(t(t-1) - (t-z)^2\big) > 0$. □

Proposition 3.9 (Reduction to the event lemma)**.** *Suppose there is a function $f \in C^2((1,z_0]) \cap C^0([1,z_0])$ with the following properties:*

(a) $f(1) = 0$, $f'(z) = o\big((z-1)^{-1}\big)$ as $z \to 1$, and $f > 0$ on $(1, z_0]$;
(b) $f'(z_0) \geq 0$;
(c) with $s := z_0 - z$ and $h := f'/f$, on $(1, z_0)$,

$$\text{(C)} \quad s\,h\,\mu \leq H, \qquad\qquad \text{(D)} \quad \tfrac{1}{2}\big(\mu' h + \mu(h' + h^2)\big) \leq s\,(Lh + L'). \tag{3.4}$$

Then every outgoing solution of (3.2) vanishes.

Proof. Cosmological side. If $L(t) > 0$, take $f_+ = 1$ on $[z_0, t]$. Then $C^{f_+} = -q_1 H \leq 0$ by Lemma 3.8, and $D^{f_+} = qL' \leq 0$ because $q \geq 0$ there. The limits of the current are computed as in [3, Lemma 4.2]:

$$\lim_{z\to t} J^{f_+}[u_+] = L(t)q(t)|B_t|^2, \qquad J^{f_+}[u_+](z_0) = -\mu(z_0)q_1 \operatorname{Re}(u'\bar{u})(z_0).$$

Since J^{f_+} is non-increasing and $L(t)q(t) > 0$, we get $\operatorname{Re}(u'\bar{u})(z_0) \leq 0$, with equality only if $B_t = 0$. If $L(t) \leq 0$, the same argument is run with the bent phase $\tilde{q} = (z - z_0)\big(q_1 + \kappa(t - z)\big)$ and the correspondingly modified $\tilde{L} = L + (\tilde{q}^2 - q^2)/\mu$, for a $\kappa > 0$ with $\tilde{L}(t) > 0$; see Proposition 3.10 below.

Event side. On $[1, z_0]$, write $q = -q_1 s$. Then $C^f \leq 0$ is equivalent to (C), and $D^f \leq 0$ is equivalent to $(\mu f')' \leq 2s(Lf)'$, which is (D). By (a), $\lim_{z\to 1} J^f[u_-] = L(1)f(1)q(1)|B_1|^2 = 0$. Monotonicity gives

$$J^f[u_-](z_0) = -\mu(z_0)q_1 f(z_0) \operatorname{Re}(u'\bar{u})(z_0) + \tfrac{1}{2}q_1\mu(z_0)f'(z_0)|u(z_0)|^2 \leq 0.$$

So $\operatorname{Re}(u'\bar{u})(z_0) \geq 0$ by (b).

Conclusion. Comparing the two sides forces $B_t = 0$. At $z = t$ the indicial roots differ by a non-integer, because $\operatorname{Im} e_t \neq 0$ in the open window, so the regular-singular argument following [3, (3.6)] gives $u = 0$. □

By Propositions 3.4, 3.5 and 3.9, the real-frequency part of Theorem 1.1 follows from the next result, which only involves the four numbers (t, z_0, A, B).

Proposition 3.10 (Cosmological side for $L(t) \leq 0$)**.** *Let w lie in the open window and suppose $L(t) \leq 0$. Then there is $\kappa > 0$ such that the bent phase $\tilde{q} = (z - z_0)\big(q_1 + \kappa(t - z)\big)$ and $\tilde{L} = L + (\tilde{q}^2 - q^2)/\mu$ satisfy $\tilde{L}(t) > 0$, $\tilde{C} := \mu'\tilde{q} - 2\mu\tilde{q}' \leq 0$ and $\tilde{D} := \tilde{q}\tilde{L}' - \kappa\mu' \leq 0$ on (z_0, t).*

With $f_+ = 1$, the current argument of [3, Lemma 4.1, §7] applies to the phase $\tilde{q}$: the derivative of the current is $\tilde{C}|u_+'|^2 + \tilde{D}|u_+|^2$, where $\tilde{D} = \tilde{q}\tilde{L}' + \frac{1}{2}(\mu\tilde{q}'')'$ and $\tilde{q}'' = -2\kappa$. Since $\tilde{q}(t) = q(t)$, the boundary limit at t is $\tilde{L}(t)q(t)|B_t|^2$. At z_0 the current equals $-\mu(z_0)\tilde{q}'(z_0) \operatorname{Re}(u'\bar{u})(z_0)$ with

$\tilde{q}'(z_0) = q_1 + \kappa(t - z_0) > 0$. So Proposition 3.10 gives the cosmological half of Proposition 3.9 when $L(t) \le 0$.

Proof. Write $\delta = z_0 - 1 > 0$ and $\theta = t - z_0 > 0$. Note $\tilde{L}(t) = L(t) + 2\kappa q_1\theta^2/(t(t-1))$.

Step 0. Since $\tilde{C}, \tilde{D}$ do not involve λ ($L' = -A/z^2 - 3$) and L increases with λ, we may replace λ by λ_{cr} (Proposition 3.7) throughout; $L(t)$ now means this value. If $L(t) < 0$ we take $\kappa = 1.05\,\kappa_{\min}$ with $\kappa_{\min} = -L(t)t(t-1)/(2q_1\theta^2)$. If $L(t) = 0$ we take $\kappa > 0$ small; every condition below is then satisfied for small κ by inspection, given (i) of Step 6.

Step 1 ($\tilde{C}$). One computes $\tilde{C} = -q_1 H - \kappa(t - z)F$, with H as in Lemma 3.8 and $F = z_0(2tz - t - z^2) - tz - z^3 + 2z^2$. Substituting $z_0 = 1 + \delta$, $z = z_0 + y$, $t = z + c$ ($\delta, y, c > 0$), the polynomial $\theta H + (t - z)F$ has only positive coefficients, so $(t - z)(-F) \le \theta H$. Hence $\tilde{C} \le -H(q_1 - \kappa\theta) \le 0$ as soon as $k\theta \le 1$, where $k := \kappa/q_1$. Write $\varpi := k\theta$.

Step 2 ($\tilde{D}$). Multiply $\tilde{D}$ by $q_1^{-3}z^2(z-1)^2 > 0$ and expand in (δ, y, c) (an exact polynomial identity). Apart from $-A\,y(z-1)^2(1+ck)/q_1^2$ and $-3yz^2(z-1)^2$, every term carries the factor k. Grouping, $\tilde{D} \le 0$ on (z_0, t) is equivalent to $A \ge P_1(y) + q_1^2 P_2(y)$ for $y \in (0, \theta)$, where, with $c = \theta - y$,

$$P_1 = \frac{z^2(gk - 3y)}{y(1 + ck)}, \quad g = y(2{+}4\delta{+}y{-}5c){-}M_0, \quad M_0 = \theta(1{+}2\delta){-}\delta(1{+}\delta), \qquad P_2 = \frac{yk\big[(2\delta z + y)(2 + ck) - kyz(z$$

Note $M_0 = \mu'(z_0)$. If $M_0 \ge 0$ then $g \le y(2 + 4\delta + \theta)$, so $P_1 \le 0$ provided

$$(S_\alpha) \quad k(2 + 4\delta + \theta) \le 3.$$

So $\tilde{D} \le 0$ follows from (S_μ): $M_0 \ge 0$, together with (S_α) and (S_β): $A \ge q_1^2 \sup_{0<y<\theta} P_2(y)$.

Step 3 (z_0-free form of ϖ). By Proposition 3.1, $\operatorname{Im} e_1 < 0$ and $\operatorname{Im} e_t < 0$ in the window. Matching exponents in Proposition 3.5 gives $q_1 = (t-1)|\operatorname{Im} e_1|/(2\delta)$ and $z_0 = 1 + (t-1)\,|\operatorname{Im} e_1|/(|\operatorname{Im} e_1| + t|\operatorname{Im} e_t|)$. Substituting into $\varpi = 1.05\,(-L(t))t(t-1)/(2q_1^2\theta)$ gives the exact identity

$$\varpi = \frac{2.1\,(-L(t))}{|\operatorname{Im} e_t|\,\big(|\operatorname{Im} e_1| + t|\operatorname{Im} e_t|\big)}.$$

Write $\tilde{\lambda} = \Lambda/3$. By Proposition 3.1, $\operatorname{Im} e_1 = 2\Xi m f_1(w)$ and $\operatorname{Im} e_t = 2\Xi m f_2(w)$, so $\varpi = \varpi_m := 2.1\,(-H_m)/\big(4\tilde{\lambda}\Xi^2 f_2(f_1 + tf_2)\big)$ with $H_m := \tilde{\lambda}L(t)/m^2$. The denominator is positive and independent of m. Moreover $H_m = \ell_2 + \ell_1 u + \ell_0 u^2$ with $u = 1/m$, where $\ell_0 = -K(1 + 7\alpha) + \tilde{\lambda}(1 - 2t) < 0$ ($K > 0$ explicit) and $\ell_2 > 0$ on the whole window (certificate, item C9). So ϖ_m is convex in u, and $\varpi_m \le \max(\varpi_2, \varpi_\infty) \le \max(\varpi_2, 0)$ for $m \ge 2$. For $m = 1$ the m-dependence of λ_{cr} differs, and ϖ_1 is treated separately. Hence $\varpi \le \bar{\varpi} := \max(\varpi_1, \varpi_2, 0)$.

Step 4 (smooth variables). Put $\gamma = A/q_1^2$, $\eta = \delta/\theta$ and $\iota = 1/\theta$. These, and ϖ_m, are rational in the window variables, independent of m (except ϖ_m), and extend smoothly to the extremal face $r_0 = r_1$. Then $(S_\alpha) \iff \varpi(1{+}4\eta{+}2\iota) \le 3$. With $y = \theta v$, $v \in (0, 1)$, and using $z - 1 = \theta(\eta + v)$ and $z = \theta(\iota + \eta + v)$, (S_β) becomes: for all $v \in (0, 1)$,

$$W(v) := \gamma(\eta{+}v)^2{-}2\varpi v P{-}\varpi^2 v E \ \ge 0, \qquad P = 2\eta(\eta{+}v){+}(2\eta{+}v)\iota, \quad E = P(1{-}v){-}v(\eta{+}v)(\eta{+}v{+}\iota).$$

Step 5 (eliminating v). Let $\varpi > 0$. Since $E \le P(1 - v)$ and $P \le 2(\eta + v)(\eta + \iota)$,

$$W(v) \ge (\eta{+}v)\big[\gamma(\eta{+}v){-}(4\varpi{+}2\varpi^2)v(\eta{+}\iota)\big] = (\eta{+}v)\big[(1{-}v)\gamma\eta{+}v\,Z(\varpi)\big], \qquad Z(\varpi) := \gamma(1{+}\eta){-}(4\varpi{+}2\varpi^2)(\eta{+}\iota).$$

The bracket is affine in v, so $W > 0$ on $(0, 1)$ if $Z(\varpi) > 0$ (recall $\gamma\eta > 0$). Both Z and $\varpi \mapsto 3 - \varpi(1 + 4\eta + 2\iota)$ are decreasing for $\varpi \ge 0$, and $\varpi \le \bar{\varpi}$. So (S_α) and (S_β) follow from: for $m = 1, 2$, either $\varpi_m \le 0$, or $Z(\varpi_m) > 0$ and $3 - \varpi_m(1 + 4\eta + 2\iota) > 0$.

Step 6 (certificates). The following are proved on the whole window by Bernstein subdivision in the variables $(r_0, 1/r_2, \tau)$, with blow-up charts at the extremal, Kerr, Nariai and ultracold

faces (Appendix A, item C9): (i) $L(t) \le 0 \Rightarrow z_f := \delta/(t-1) < \frac{1}{2}$; (ii) $\varpi_m \le 1$ for $m = 1, 2$; (iii) for $m = 1, 2$: $\varpi_m \le 0$ or $3 - \varpi_m(1 + 4\eta + 2\iota) > 0$; (iv) for $m = 1, 2$: $\varpi_m \le 0$ or $Z(\varpi_m) > 0$; (v) $f_1 < 0$, $f_2 < 0$ and $\ell_2 > 0$. From (i), $z_0 < (1+t)/2$; since μ' is a concave quadratic with $\mu'(1) = t - 1 > 0$ and $\mu'((1+t)/2) = (t-1)^2/4 > 0$, we get $M_0 = \mu'(z_0) > 0$. By Steps 1–5, (i)–(v) give $\tilde{C} \le 0$, $\tilde{D} \le 0$ and $\tilde{L}(t) > 0$. □

Theorem 3.11 (Event lemma)**.** *For all $t > 1$, $z_0 \in (1, t)$ and $A > 0$ with $L(z_0) \ge A/(2z_0) + 2t$, there exists f satisfying (a)–(c) of Proposition 3.9.*

4. Proof of the event lemma

In this section and in Proposition 3.9, s denotes $z_0 - z$; the spin plays no role here.

In terms of f, condition (C) reads $f'/f \le h_C := H/(s\mu) = 2/s + \mu'/\mu$. Condition (D) reads $\widetilde{D}f := (\mu f')' - 2s(Lf)' \le 0$. Put $x_0 = z_0 - 1$ and

$$p = \frac{2x_0 L(1)}{t-1}.$$

We distinguish three regimes:

$$\text{R1: } p \le 1; \qquad \text{R2: } p > 1 \text{ and } 2x_0^2\Big(\frac{A}{z_0} + 3\Big) < t - 1; \qquad \text{R3: the rest.}$$

4.1. Regimes R1 and R2: an explicit solution.

Proposition 4.1. *In R1 and R2, the conclusion of Theorem 3.11 holds.*

Proof. Let $\tilde{L} = L$ in R1, and $\tilde{L} = L - \kappa$ with $\kappa = L(1) - \frac{t-1}{2x_0} > 0$ in R2. Let f solve $(\mu f')' = 2s(\tilde{L}f)'$ with $f(1) = 0$, that is,

$$\mu f' = 2s\tilde{L}f + 2\int_1^z \tilde{L}f.$$

At $z = 1$ this is a regular-singular equation, and $f \sim (z-1)^{\tilde{p}}$ with $\tilde{p} = 2x_0\tilde{L}(1)/(t-1) \in (0, 1]$. So (a) holds. In R2 we have $\tilde{L}(z_0) > 0$, since $L(1) - L(z_0) = Ax_0/z_0 + 3x_0$ and this is exactly the R2 condition. Thus $\tilde{L} > 0$ on $[1, z_0]$, so $\mu f' > 0$ and f is positive and increasing; in particular (b) holds.

For (D): $(\mu f')' = 2s(\tilde{L}f)' \le 2s(Lf)'$, because $((L - \tilde{L})f)' = \kappa f' \ge 0$.

For (C): $\tilde{L}$ is decreasing, so

$$s\mu f' \le 2s\tilde{L}(1)f\,(s + z - 1) = 2sx_0\tilde{L}(1)f \le s(t-1)f.$$

It remains to show that $H \ge s(t-1)$. With $z = 1 + x$,

$$H - s(t-1) = x\big[2(1+x)(t-1-x) + s(2t-4-3x)\big].$$

The bracket is affine in s. It is nonnegative at $s = 0$, and at $s = t - 1 - x$ it equals $(t - 1 - x)(2t - 2 - x) \ge 0$. □

4.2. Regime R3: saturate (C), then solve (D).

Proposition 4.2. *In R3, the conclusion of Theorem 3.11 holds.*

The construction follows [3, §4.3.3]. Since $h_C = (\log F_C)'$ with

$$F_C := \mu/s^2,$$

we first take $f = F_C$, for which (C) holds with equality. Condition (D) then becomes $d := \widetilde{D}F_C/(2F_C) \le 0$. A direct computation gives $\widetilde{D}F_C = N/s^4$, where in the variable $S = z_0 - z \in (0, x_0)$

$$N = 6\mu_0^2 - 6\mu_0\mu_0' S + c_2 S^2 + c_3 S^3 + 2(B + 4t - 15z_0 + 4)S^5 + 15S^6. \tag{4.1}$$

Here $\mu_0 = \mu(z_0)$, $\mu_0' = \mu'(z_0)$, and c_2, c_3 are affine in (A, B) with explicit polynomial coefficients in (t, z_0). The coefficient of S^4 vanishes.

Lemma 4.3 (One sign change)**.** *In R3, N has exactly one zero $\hat{S} \in (0, x_0)$. Hence $d < 0$ near $z = 1$, $d > 0$ near z_0, and d changes sign exactly once, at $\hat{z} = z_0 - \hat{S}$.*

Proof. Write $B = 3z_0 - A/z_0 + \beta$ with $\beta = L(z_0) > 0$. Substitute $S = x_0 v/(1+v)$, $v \in (0, \infty)$, clear denominators, and let $T_0, \dots, T_6$ be the coefficients of the resulting sextic in v. $T_0 = 6z_0^3(z_0 - t)^2 x_0^2 > 0$ and $T_1 = 6z_0^2(z_0 - t)x_0^2(3z_0^2 + 2z_0 - 4tz_0 - t) > 0$.

$T_6 = z_0 N|_{z=1} = -z_0(t-1)^2 x_0^2(p-1) < 0$ in R3, by an exact factorization.

For $k = 3, 4, 5$, T_k is affine in (A, β) with strictly negative coefficients, and T_{k-1} is affine with nonpositive coefficients. In R3 the admissible set is the quadrant $\{A \ge a_0, \beta \ge 0\}$ with $a_0 = \max(0, z_0((t-1)/(2x_0^2) - 3))$, since the line $p = 1$ passes through its corner. So $\{T_k \ge 0\}$ meets the quadrant in a triangle at the corner, and $T_{k-1} > 0$ on it if two explicit combinations g_1, g_2 of corner values and coefficients are positive. These are rational functions of (t, z_0); we certify $g_1, g_2 > 0$ unconditionally for $k - 1 = 2, 3, 4$ (Appendix A, item C3). Hence $T_k \le 0 \Rightarrow T_{k+1} < 0$ for $k = 2, \dots, 5$, so the sign pattern changes exactly once. By Descartes' rule, N has exactly one root in $(0, x_0)$. □

Proof of Proposition 4.2. Define $f = F_C$ on $(1, \hat{z}]$, and on $[\hat{z}, z_0]$ let f solve $\widetilde{D}f = 0$ with $(f, f')(\hat{z}) = (F_C, F_C')(\hat{z})$. Since $d(\hat{z}) = 0$, f is C^2. Condition (a) holds because $F_C \sim (t-1)(z-1)/x_0^2$ as $z \to 1$. Condition (D) holds on $(1, \hat{z}]$ by Lemma 4.3, and with equality on $[\hat{z}, z_0]$. Condition (C) on $[\hat{z}, z_0]$ follows from the Wronskian argument: with $p_* = \mu \exp(-\int 2sL/\mu)$, the function $W = p_*(F_C f' - F_C' f)$ is non-increasing where $d \ge 0$ and $W(\hat{z}) = 0$.

The seam. It remains to show that $f'(z_0) > 0$. Put $w = \mu f'/f$. It satisfies the Riccati equation

$$\mu w' = R(w) := -w^2 + 2sLw - 2\mu s|L'|, \qquad w(\hat{z}) = \hat{w} := \hat{H}/\hat{s} > 0, \tag{4.2}$$

and w can cross 0 only downward. A function k with $\mu k' \le R(k)$ in the Dini sense and $k(\hat{z}) \le \hat{w}$ stays below w. We use $k = \min(\hat{w}, sL)$.

(i) sL is a strict subsolution on $(1, z_0)$. Since $(sL)' = -L - s|L'|$, $R(sL) - \mu(sL)' = g := s^2L^2 + \mu(L - s|L'|)$, and $L - s|L'| = \ell - As^2/(z^2 z_0)$ exactly, where $\ell = L(z_0)$. As L is decreasing, (3.3) gives $L \ge \ell \ge A/(2z_0) + 2t$, hence $L^2 \ge 4At/z_0 \ge \mu A/(z^2 z_0)$ (because $\mu < tz^2$). Therefore $g \ge \mu\ell > 0$.

(ii) The constant $\hat{w}$ is a subsolution until sL falls below it. On the interval where $sL \ge \hat{w}$ we have $\hat{w} \le sL < k_+$, where $k_\pm = sL \pm \sqrt{s^2L^2 - 2\mu s|L'|}$ are the roots of R. So $R(\hat{w}) > 0$ follows from $\hat{w} > k_- = 2\mu s|L'|/k_+$, which in turn follows from

$$\hat{w}L(z) > 2\mu(z)|L'(z)| \qquad (z \in [\hat{z}, z_0)). \tag{4.3}$$

To prove (4.3), note that $Q := \hat{w}L - 2\mu|L'|$ is affine and increasing in B. At the true value $B = B_*(\hat{s})$ (the root condition $N(\hat{s}) = 0$), (3.3) gives $B_* \ge B_m(A) := 3z_0 - A/(2z_0) + 2t$. Since N is decreasing in B, this is the condition $g_N := N(\hat{s}; A, B_m) \ge 0$, and $Q(B_*) = Q(B_m) + \hat{w}\, g_N/|\partial_B N|$. Hence it suffices to prove: for $t > z_0 > 1$, $A \ge 0$, $\hat{s} \in (0, x_0)$ and $z \in [z_0 - \hat{s}, z_0)$,

$$N(\hat{s}; A, B_m(A)) \ge 0 \implies Q(B_m) > 0.$$

Both sides are affine in A, and the coefficient of A in N is negative, so the admissible A form a segment $[0, A_*]$. It therefore suffices to check $A = 0$ and $A = A_*$. After compactifying $(z_0, t, \hat{s}, z)$ to $(0,1)^4$, these are two polynomial implications of degree at most $(10, 3, 10, 4)$, plus the corresponding statements at $z = \hat{z}$. We certify them by an exact Bernstein covering (Appendix A, item C4).

By (i) and (ii), $k = \min(\hat{w}, sL)$ is a subsolution on $[\hat{z}, z_0)$ with $k(\hat{z}) \le \hat{w}$. Hence $w > 0$ there. Since $g \ge \mu\ell$ stays positive up to z_0, we get $w(z_0) > 0$, i.e. $f'(z_0) > 0$. So (b) holds, and $f > 0$. □

This completes the proof of Theorem 3.11, and with it Theorem 1.1 for real $\sigma \neq 0$.

5. Zero frequency

Proposition 5.1. *There are no nontrivial outgoing modes at $\sigma = 0$, for $s = \pm 2$.*

Proof. Case $am \neq 0$. Then $\kappa_1, \kappa_2, \kappa_3$ are nonzero and $w = 0 < w_2$ lies outside the window; the proof of Corollary 3.2 applies verbatim. If $a = 0$, the radial equation at $\sigma = 0$ does not involve m, and the argument below applies with $\lambda = \ell(\ell+1) - 4 \ge 2$.

Case $m = 0$ or $a = 0$, $s = -2$. The radial equation is $(\Delta_r^{-1} R')' + V\Delta_r^{-2} R = 0$, with $V = -2\Lambda r^2 - 2(1-\alpha) - \lambda$. Put $g = \Delta_r^2 R''''$. The Teukolsky–Starobinsky identity at $K = 0$ shows that g solves the same equation. Moreover g vanishes like Δ_r^2 at both horizons, so the energy identity $\int \Delta_r^{-1}|g'|^2 = \int V\Delta_r^{-2}|g|^2 \le 0$ gives $g = 0$. Thus R is a cubic polynomial. Inserting a cubic into the equation gives a 4×4 linear system for its coefficients, with determinant $[\lambda^2 + 6(1-\alpha)\lambda + 8 - 4\alpha + 8\alpha^2]^2$. Both roots of the quadratic are negative, and $\lambda \ge \lambda_{\mathrm{cr}} = 2 - 4\alpha > 0$ at $m = \xi = 0$ by Proposition 3.7 and Lemma 2.1. Hence the determinant is positive and $R = 0$.

Case $m = 0$ or $a = 0$, $s = +2$. The function $\Delta_r^2 R_{+2}$ solves the $s = -2$ equation, so we may repeat the argument.

The two identities used are exact symbolic identities (Appendix A, item C5). □

6. The upper half plane

6.1. Fredholm theory.

Lemma 6.1 (Regular frame)**.** *Let $\chi \in C^\infty(\mathbb{R})$ equal 1 near r_1 and 0 near r_2, and put $b(r) = \chi(r) + (1 - \chi(r))/\Delta_r$ on (r_1, r_2). Let (l_K, n_K, m_K) be the Kinnersley tetrad. Then*

$$l = b\,\Delta_r\, l_K, \qquad n = (b\,\Delta_r)^{-1} n_K, \qquad m = m_K$$

extends to a stationary null tetrad that is smooth on $M = [r_1 - \delta, r_2 + \delta] \times S^2$, away from the usual phase ambiguity of m on the axis. In this tetrad, the GHP Teukolsky operator of spin $s = \pm 2$ is a stationary second-order operator with smooth coefficients on sections of a complex line bundle over M, with principal symbol $\frac{1}{2}G(\zeta, \zeta)$, where G is the dual metric. Its outgoing modes are exactly the modes of Lemma 2.2.

Proof. Near r_1, in ingoing coordinates $(v, r, \tilde{\varphi})$,

$$\Delta_r l_K = 2\Xi(r^2 + a^2)\partial_v + 2\Xi a\partial_{\tilde{\varphi}} + \Delta_r \partial_r, \qquad \Delta_r^{-1} n_K = -\partial_r/(2\varrho^2).$$

Both are smooth and nonvanishing. Near r_2, in outgoing coordinates, $l_K = \partial_r + (\text{angular})$ and n_K are already smooth. The factor $b\Delta_r$ is smooth and positive on (r_1, r_2), and equals Δ_r near r_1 and 1 near r_2. The GHP operator is covariant, so its coefficients in this tetrad (spin coefficients and directional derivatives) are smooth. Its principal part is $\text{þ}\text{þ}' - \eth\eth'$, whose symbol is $\zeta(l)\zeta(n) - \zeta(m)\zeta(\bar{m}) = \frac{1}{2}G(\zeta, \zeta)$. The last statement is the definition of outgoing modes, together with Lemma 2.2. □

Fix $m \in \mathbb{Z}$, and let $P_b(\sigma)$ be the spectral family of this operator on $\ker(\partial_{\varphi_*} - im)$, at parameters $b = (\Lambda, M, a)$. The dynamical hypotheses of [5, §5.1] concern only the principal symbol G (radial sets at the horizons, and the global null-bicharacteristic flow). They were verified in the full subextremal range by Petersen–Vasy [10, 9]; see [4, Remark 2.12]. The subprincipal part enters only through the threshold regularity at the radial sets. The trapping hypothesis of [5, §5.1] is needed only for high-energy estimates near the real axis, not for the Fredholm property [5, Theorem 5.4 and the remark in its proof]. We therefore have:

- (F1) $P_b(\sigma)$ is a holomorphic family of Fredholm operators of index 0 on $\operatorname{Im}\sigma > -c$ [5, Theorem 5.4]. Its non-invertibility set is discrete, and it is the set of quasinormal mode (QNM) frequencies.
- (F2) There is C_0, uniform for b near b_0, such that $P_b(\sigma)$ is invertible for $\operatorname{Im}\sigma > C_0$. This follows from the global energy estimate on growing function spaces, which uses only hyperbolicity and holds uniformly in b [5, §5.1.2].
- (F3) The non-invertibility set is closed in (b, σ), and QNMs move continuously, with constant total rank in small discs [5, Proposition 5.11]. That proof uses only the Fredholm estimates [5, (5.26)] on a precompact set of σ, which hold uniformly in b.

What is *not* available for Teukolsky in the full subextremal range is the high-energy estimate bounding $|\operatorname{Re}\sigma|$. We obtain it sector by sector from the gauge-fixed Einstein operator, for which it is proved in [4].

6.2. **Metric reconstruction.** We use the Geroch–Held–Penrose (GHP) formalism with $\text{þ}' = \Delta - p\gamma - q\bar{\gamma}$ and $\text{ð}' = \bar{\delta} - p\alpha - q\bar{\beta}$ [1]. Let $\mathcal{O}$ be the GHP Teukolsky operator for Ψ_0, and $\mathcal{O}'$ its prime. On any type D Λ-vacuum the Bianchi identities give $\text{þ}\Psi_2 = 3\rho\Psi_2$, $\text{ð}\Psi_2 = 3\tau\Psi_2$ and their primes; there are no Λ-terms. With $P := \Psi_2^{-4/3}$ (GHP type $(0,0)$), and using the formal adjoint rules $\text{þ}^\dagger = -(\text{þ} - \rho - \bar{\rho})$ and $\text{ð}^\dagger = -(\text{ð} - \tau - \bar{\tau}')$, a direct computation gives

$$\mathcal{O}'^\dagger \circ P = P \circ \mathcal{O}. \tag{6.1}$$

Proof of (6.1). Write $\mathcal{O} = (\text{þ} - 4\rho - \bar{\rho})(\text{þ}' - \rho') - (\text{ð} - 4\tau - \bar{\tau}')(\text{ð}' - \tau') + V_0$ and $\mathcal{O}'$ for its prime, where the zeroth-order term V_0 (a constant multiple of Ψ_2, plus a Λ-term) is prime-invariant and real. Using $(AB)^\dagger = B^\dagger A^\dagger$ and the adjoint rules,

$$(\text{þ} - \rho)^\dagger = -(\text{þ} - \bar{\rho}), \qquad (\text{þ}' - 4\rho' - \bar{\rho}')^\dagger = -(\text{þ}' + 3\rho'),$$

and similarly for ð, with $(\rho, \bar{\rho}, \rho')$ replaced by $(\tau, \bar{\tau}', \tau')$. Hence

$$\mathcal{O}'^\dagger = (\text{þ} - \bar{\rho})(\text{þ}' + 3\rho') - (\text{ð} - \bar{\tau}')(\text{ð}' + 3\tau') + V_0.$$

From $\text{þ}\Psi_2 = 3\rho\Psi_2$, $\text{þ}'\Psi_2 = 3\rho'\Psi_2$, $\text{ð}\Psi_2 = 3\tau\Psi_2$ and $\text{ð}'\Psi_2 = 3\tau'\Psi_2$ we get $\text{þ}P = -4\rho P$, $\text{þ}'P = -4\rho' P$, $\text{ð}P = -4\tau P$ and $\text{ð}'P = -4\tau' P$. Therefore

$$(\text{þ} - \bar{\rho})(\text{þ}' + 3\rho')(PX) = (\text{þ} - \bar{\rho})\big(P(\text{þ}' - \rho')X\big) = P(\text{þ} - 4\rho - \bar{\rho})(\text{þ}' - \rho')X,$$

the same computation applies to the angular term, and V_0 commutes with P. This proves (6.1) as an operator identity.

Wald's argument [13], which uses only the decoupling identity $\mathcal{S}'\mathcal{E} = \mathcal{O}'\mathcal{T}'$ and the self-adjointness of the linearized Einstein operator $\mathcal{E}$, then gives the following.

Lemma 6.2 (Reconstruction). *If $\mathcal{O}\Psi = 0$, then $h_\Psi := \mathcal{S}'^\dagger(P\Psi)$ solves $\mathcal{E}h_\Psi = 0$. Moreover, the complex conjugate $\tilde{h} := \overline{h_\Psi}$ satisfies $\Psi_4[\tilde{h}] = c_0(\text{þ}')^4(\bar{P}\bar{\Psi})$ for a universal constant $c_0 \neq 0$ ($c_0 = \frac{1}{2}$ in the normalization of* [7]*).*

Proof. The first claim is Wald's argument combined with (6.1): from $\mathcal{S}'\mathcal{E} = \mathcal{O}'\mathcal{T}'$ and the self-adjointness of $\mathcal{E}$ we get $\mathcal{E}\mathcal{S}'^\dagger = \mathcal{T}'^\dagger\mathcal{O}'^\dagger$, and $\mathcal{O}'^\dagger(P\Psi) = P\mathcal{O}\Psi = 0$.

For the second claim we use the convention $h_{xy} := h_{ab}x^a y^b$. The operator $\mathcal{S}'$ only involves the components T_{nn}, $T_{n\bar{m}}$ and $T_{\bar{m}\bar{m}}$. So $\mathcal{S}'^{\dagger}$ produces a tensor of the form $h_\Psi = X\, n\otimes n + Y\, n\odot\bar{m} + Z\,\bar{m}\otimes\bar{m}$ (indices lowered), whose only nonzero components are h_{ll}, h_{lm} and h_{mm}. By GHP type counting, the linearized Ψ_4 depends only on h_{nn}, $h_{n\bar{m}}$ and $h_{\bar{m}\bar{m}}$. Hence $\Psi_4[h_\Psi] = 0$. For $\tilde{h} = \overline{h_\Psi}$ the only nonzero components are h_{ll}, $h_{l\bar{m}}$ and $h_{\bar{m}\bar{m}}$, so $\Psi_4[\tilde{h}]$ comes from $\tilde{h}_{\bar{m}\bar{m}} = \overline{h_{mm}}$ alone. The resulting expression involves only $\text{þ}'$-derivatives and the coefficients $\rho', \bar{\rho}'$. Its evaluation uses only the Ricci identities $\text{þ}'\rho' = \rho'^2$ and $\text{þ}'\bar{\rho}' = \bar{\rho}'^2$, so no commutator and no Λ-term enters. The algebra is therefore literally that of the Kerr case [13, 7], which gives $\Psi_4[\tilde{h}] = c_0(\text{þ}')^4(\bar{P}\bar{\Psi})$. Only $c_0 \neq 0$ is used below. □

6.3. **The kernel lemma.** Let $\kappa_2^\sharp = |\Delta_r'(r_2)|/(2\Xi\rho_2)$ be the normalized surface gravity of the cosmological horizon, and define the *cosmological ladder*

$$\mathcal{Z}_m = \{ m\Omega_2 + iN\kappa_2^\sharp :\ N \in \{1,2,3,\dots\}\}.$$

Lemma 6.3 (Kernel lemma)**.** *Let* $\operatorname{Im}\sigma > 0$ *and* $\sigma \notin \mathcal{Z}_m$*. If* $F = e^{-i\sigma t_* + im\varphi_*}v$ *is a mode of GHP type* $(0,4)$ *or* $(4,0)$*, regular at the future cosmological horizon, with* $(\text{þ}')^4 F = 0$*, then* $F = 0$*.*

Proof. In the tetrad of Lemma 6.1, $\text{þ}'$ differentiates only along n, which has no ∂_θ-component. So for each fixed θ, $(\text{þ}')^4 F = 0$ is a fourth-order linear ODE in r, the composition of four first-order operators

$$n(\cdot) - (p_i + q_i)\gamma, \qquad p_i + q_i \in \{4,2,0,-2\}.$$

On modes, $n = -(\Delta_r/2\varrho^2)\big(\partial_r + iK/\Delta_r\big)$ in the Kinnersley normalization, and $\gamma = \mu + \Delta_r'/(4\varrho^2)$ with μ vanishing like Δ_r. Hence each first-order factor has kernel spanned by $\exp(-i\int K/\Delta_r)$ times Δ_r^k times an analytic function, with $k \in \frac{1}{2}(p_i+q_i) + \mathbb{Z} \subset \mathbb{Z}$. Every local solution near r_2 is therefore a finite sum of terms $(r_2 - r)^{-\kappa_2 + k}\log^j(r_2 - r)$ times analytic functions, with $k \in \mathbb{Z}$.

Regularity at the future cosmological horizon is the branch $(r_2 - r)^{\kappa_2}C^\infty$ (Lemma 2.2). So a nonzero F requires $2\kappa_2 \in \mathbb{Z}$. For $\operatorname{Im}\sigma > 0$ we have $\operatorname{Re}(2\kappa_2) = 2\Xi\rho_2 \operatorname{Im}\sigma/|\Delta_r'(r_2)| > 0$ and $\operatorname{Im}(2\kappa_2) = 2\Xi(\rho_2 \operatorname{Re}\sigma - am)/\Delta_r'(r_2)$. Hence $\operatorname{Re}\sigma = m\Omega_2$ and $\operatorname{Im}\sigma = N\kappa_2^\sharp$ with $N = 2\kappa_2 \in \{1,2,3,\dots\}$. Finally, F vanishes identically once it vanishes near r_2, by uniqueness for the ODE. □

Proposition 6.4 (Transfer)**.** *Let* Ψ *be an outgoing* $s = +2$ *Teukolsky mode at frequency* σ *in the sector* m*, with* $\operatorname{Im}\sigma > 0$ *and* $\sigma \notin \mathcal{Z}_m$*. Then* $\tilde{h}$ *from Lemma 6.2 is a smooth outgoing mode of the linearized Einstein equation, at frequency* $-\bar{\sigma}$ *in the sector* $-m$*, with* $\Psi_4[\tilde{h}] \neq 0$*. The case* $s = -2$ *is the same with the roles of* l *and* n*, and of the two horizons, exchanged: one uses the ingoing gauge and the ladder* $\mathcal{Z}_m^e$ *of the event horizon.*

Proof. Smoothness. $\mathcal{S}'^{\dagger}$ is GHP-covariant. Near each future horizon it may therefore be evaluated in a principal tetrad that is smooth there. In such a tetrad, outgoing modes are smooth, so $\tilde{h}$ is smooth.

Nonvanishing of Ψ_4. The function $\bar{P}\bar{\Psi}$ is regular at both horizons, and its frequency $-\bar{\sigma}$ lies off $\mathcal{Z}_{-m} = -\overline{\mathcal{Z}_m}$. By Lemma 6.3, $(\text{þ}')^4(\bar{P}\bar{\Psi}) \neq 0$. □

6.4. **Compactness, sector by sector.**

Lemma 6.5. *Let* b_0 *be subextremal and* $m \in \mathbb{Z}$*. There are a neighborhood* $U \ni b_0$ *and* $R_m < \infty$ *such that, for all* $b \in U$*, every Teukolsky QNM of spin* ± 2 *in the sector* m *with* $\operatorname{Im}\sigma \geq 0$ *satisfies* $|\sigma| \leq R_m$*.*

Proof. Let L_b be the linearized gauge-fixed Einstein operator of [4, (23)], with the constraint damping fixed at b_0. By [4, Remark 2.12 and §2.5], the operators L_b, $b \in U$, form a continuous

family of stationary, principally scalar operators satisfying the assumptions of [5, §5.1], including the trapping condition (by [4, Theorem 3.1]). By [5, §5.1.2], these assumptions then hold uniformly near b_0, and the high-energy estimates [5, (5.7), (5.8)] hold with constants uniform in $b \in U$. In particular there is R_0 such that every QNM of L_b with $\operatorname{Im} \geq 0$ has $|\sigma_T| \leq R_0$ for all $b \in U$. Here σ_T is the T-frequency, which differs from the ∂_t-frequency by $m'\Omega_T$ on the sector m'.

Let Ψ be a QNM in the sector m. By Theorem 1.1 on the real axis and (F2), $0 < \operatorname{Im}\sigma \leq C_0$, so ladder points satisfy $|\sigma| \leq |m| \max(\Omega_1, \Omega_2) + C_0$. Otherwise Proposition 6.4 gives $\tilde{h}$ at $-\bar{\sigma}$ with $\Psi_4[\tilde{h}] \neq 0$, and we gauge-fix it. The map $\omega \mapsto D\Upsilon(\delta^*\omega)$ is a principally scalar wave operator on 1-forms, hence Fredholm of index 0 (F1). If it is invertible at $-\bar{\sigma}$, then $h' = \tilde{h} + \delta^*\omega$ with $D\Upsilon(h') = 0$ is a QNM of L_b, nonzero since $\Psi_4[h'] = \Psi_4[\tilde{h}]$. If not, a nonzero ω in its kernel gives the QNM $\delta^*\omega$ of L_b, nonzero because KdS has no Killing modes with $\operatorname{Im} \neq 0$. Either way L_b has a QNM at T-frequency $-\bar{\sigma} + m\Omega_T$, so $|\sigma| \leq R_0 + |m|\Omega_T$. Take $R_m = \max\big(R_0 + |m|\Omega_T,\ |m| \max(\Omega_1, \Omega_2) + C_0\big)$. □

6.5. **The base case.**

Lemma 6.6 (Schwarzschild–de Sitter)**.** *For all but countably many values of* $\Lambda M^2 \in (0, 1/9)$*, SdS has no Teukolsky QNMs of spin* ± 2 *with* $\operatorname{Im}\sigma > 0$*.*

Proof. At $a = 0$ the angular operator is independent of σ and Λ, so modes separate for every $\sigma \in \mathbb{C}$, with $\lambda_\ell = \ell(\ell+1) - 4$, $\ell \geq 2$. In the Kinnersley tetrad, with $D^\dagger = \partial_r + i\sigma r^2/\Delta_r$ and $D = \partial_r - i\sigma r^2/\Delta_r$, one has (item C8)

(6.2)

$$(\text{þ}')^4(r^4 R) = \frac{\Delta_r^2 (D^\dagger)^4 \Delta_r^2 R}{16 r^4}, \qquad D^4 \Delta_r^2 (D^\dagger)^4 \Delta_r^2 R = \mathcal{C}_\ell(\sigma) R, \qquad \mathcal{C}_\ell(\sigma) = [(\lambda_\ell+2)(\lambda_\ell+4)]^2 + 144 M^2 \sigma^2,$$

the second for every solution of the $s = +2$ radial equation. As $P \propto r^4$, $(\text{þ}')^4(P\Psi) = 0$ forces $\mathcal{C}_\ell(\sigma) = 0$, i.e. $\sigma = i(\lambda_\ell + 2)(\lambda_\ell + 4)/(12M)$, and by Lemma 6.3 also $\sigma = iN\kappa_2^\sharp$ with $N \geq 1$. Since $M\kappa_2^\sharp$ is a nonconstant analytic function of ΛM^2, the two coincide only for countably many ΛM^2. Off these, every QNM with $\operatorname{Im}\sigma > 0$ yields, by Lemma 6.2, a linearized Einstein mode with $\operatorname{Im} > 0$ and $\Psi_4 \neq 0$; but on SdS such modes are pure gauge [5, §7], so $\Psi_4 = 0$. For $s = -2$ use the ingoing gauge, þ^4 and the event horizon. □

6.6. **Conclusion.**

Proof of Theorem 1.1 for $\operatorname{Im}\sigma > 0$. Fix a subextremal b_1 and $m \in \mathbb{Z}$. Choose an SdS point b_* as in Lemma 6.6, and a continuous path $p : [0,1] \to \{\text{subextremal parameters}\}$ with $p(0) = b_*$ and $p(1) = b_1$. For example, scale r_0 to 0 with r_1, r_2 fixed, as in [3, §6], and then move within the SdS family. Let S be the set of $s \in [0,1]$ for which the sector m has a QNM with $\operatorname{Im}\sigma > 0$. S is open, by (F3).

S is closed. Let $s_j \to s_0$ with $s_j \in S$ and QNMs σ_j. By Lemma 6.5 and (F2), the σ_j lie in the compact set $\{0 \leq \operatorname{Im}\sigma \leq C_0,\ |\sigma| \leq R_m\}$. A subsequence converges to some σ_0, which is a QNM at $p(s_0)$ by (F3). Now $\sigma_0 \neq 0$, because $P(0)$ is invertible by Proposition 5.1 and invertibility is open. And $\sigma_0 \notin \mathbb{R}$ by §4 and Corollary 3.2. Hence $\operatorname{Im}\sigma_0 > 0$.

$0 \notin S$, by Lemma 6.6. Here (F2), (F3) and Lemma 6.5 are applied in finitely many neighborhoods U covering the compact path. Hence $S = \emptyset$. Generalized modes reduce to modes, since their top coefficient is a mode. □

7. Ungauged linearized Einstein mode stability

Throughout this section $a \neq 0$, and $\Omega_T = a/(r_2^2 + a^2)$, so that $T = \partial_t + \Omega_T \partial_\varphi$ in Boyer–Lindquist coordinates. Let h be as in Theorem 1.2. Its Fourier components in φ_* are generalized $(\partial_t, \partial_\varphi)$-modes of type $(\sigma + m\Omega_T, m)$, whose imaginary part is $\operatorname{Im}\sigma \geq 0$. The linearized Teukolsky scalars $\delta\Psi_0$ and $\delta\Psi_4$ are gauge and tetrad invariant on type D, and they are outgoing generalized Teukolsky modes. By Theorem 1.1, applied to each Fourier component of h and of $\bar{h}$, the scalars $\delta\Psi_0$ and $\delta\Psi_4$ vanish on the exterior $r_1 < r < r_2$. The next lemma extends this to all of M:

$$\delta\Psi_0 = \delta\Psi_4 = 0 \quad \text{on } M.$$

Lemma 7.1 (Collars)**.** *Let w be a smooth generalized mode on M of a stationary, principally scalar operator with smooth coefficients and principal symbol $\frac{1}{2}G$, for instance the Teukolsky operator of Lemma 6.1. If $w = 0$ on $r_1 < r < r_2$, then $w = 0$ on M.*

Proof. Consider the collar $r_1 - \delta < r < r_1$. There dr is timelike, and r decreases towards the future. Let p lie in the collar. Every past-directed causal curve from p has strictly increasing r. The set $D_p = J^-(p) \cap \{r \leq r_1\}$ is compact, since in ingoing coordinates the past cone of p meets the event horizon in a compact set. Its boundary consists of a compact piece of the event horizon and a piece of the past light cone of p. Since w is smooth and vanishes for $r > r_1$, it vanishes to infinite order on the horizon. The standard energy estimate on D_p involves fluxes through a null boundary that are tangential and hence vanish, and fluxes through the cone of p that are nonnegative [2, §23.2]. It gives $w = 0$ near p. The collar $r_2 < r < r_2 + \delta$ is treated in the same way, with time reversed relative to r and the cosmological horizon in place of the event horizon. □

7.1. Linearized Goldberg–Sachs.

Lemma 7.2. *After an infinitesimal null rotation of the tetrad, $\delta\Psi_1 = \delta\Psi_3 = 0$, and then $\delta\kappa = \delta\sigma = \delta\kappa' = \delta\sigma' = 0$.*

Proof. To first order, a null rotation about n with parameter c shifts $\delta\Psi_1$ by $3c\Psi_2$ and fixes $\delta\Psi_0$ and $\delta\Psi_2$; the analogous statement holds about l. Since $\Psi_2 \neq 0$, the choice is algebraic. The Λ-vacuum Bianchi identities contain no Λ-terms. Linearizing

$$\text{Þ}\Psi_1 - \eth'\Psi_0 = -\tau'\Psi_0 + 4\rho\Psi_1 - 3\kappa\Psi_2, \qquad \text{Þ}'\Psi_0 - \eth\Psi_1 = \rho'\Psi_0 - 4\tau\Psi_1 + 3\sigma\Psi_2$$

gives $\delta\kappa = \delta\sigma = 0$; priming gives the rest. □

All perturbed quantities below are local linear functions of h. They are boost-spin invariant and commute with $\mathcal{L}_{\partial_t}$ and $\mathcal{L}_{\partial_\varphi}$.

7.2. The Killing-spinor field.

On a type D Λ-vacuum with $\kappa = \sigma = \kappa' = \sigma' = 0$, the following identities hold, together with their primes:

$$\text{Þ}\rho = \rho^2, \quad \eth\rho = (\rho - \bar{\rho})\tau, \quad \text{Þ}\tau = (\tau - \bar{\tau}')\rho, \quad \eth\tau = \tau^2,$$
$$\text{Þ}\rho' - \text{Þ}'\rho = \tau\bar{\tau} - \tau'\bar{\tau}', \quad \eth\tau' - \eth'\tau = \rho\bar{\rho}' - \bar{\rho}\rho', \quad \text{Þ}\tau' = \eth'\rho, \quad \text{Þ}'\tau = \eth\rho'.$$

They follow from the GHP Ricci identities and commutators. We verified them exactly on KdS, and on the accelerating NUT Plebański–Demiański metric, which has $\tau \neq -\bar{\tau}'$ (Appendix A, item C6).

Let $\phi = \Psi_2^{-1/3}$ (normalized so that $\phi = r - ia\cos\theta$ on KdS), and put

$$A = \phi\rho, \quad B = -\phi\rho', \quad C = \phi\tau, \quad D = -\phi\tau', \qquad \xi = Bl + An - Dm - C\bar{m}.$$

Lemma 7.3. *The ten Killing equations for ξ are algebraic consequences of the identities above. On KdS, $\xi = -\Xi\,\partial_t$.*

Linearizing Lemma 7.3 gives $\mathcal{L}_\xi h + \mathcal{L}_{\delta\xi} g = 0$, where $\delta\xi = \delta\xi[h]$.

Lemma 7.4 (Reality)**.** *Put* $J[h] = \big(\delta\xi[h] - \overline{\delta\xi[\bar{h}]}\big)/(2i)$. *Then* $\mathcal{L}_{J[h]} g = 0$. *If* $(\sigma, m) \neq (0,0)$, *then* $J[h] = 0$.

Proof. For real h, $J[h] = \operatorname{Im}\delta\xi$, and the imaginary part of the linearized Killing equation reads $\mathcal{L}_J g = 0$; by complex linearity this holds for all h.

The local Killing algebra of KdS with $a \neq 0$ is $\operatorname{span}\{\partial_t, \partial_\varphi\}$ on every open set: a Killing field preserves Ψ_2, hence r and $a\cos\theta$. It is therefore tangent to the (t,φ)-orbits, and the isotropy is trivial because it must fix ∇r, $\nabla(a\cos\theta)$ and the principal null directions.

Moreover, J is complex linear and commutes with $\mathcal{L}_{\partial_t}$ and $\mathcal{L}_{\partial_\varphi}$. For a mode of type $(\sigma, m) \neq (0,0)$, it is therefore a constant combination of $\partial_t, \partial_\varphi$ of that type, hence zero. □

Remark 7.5. The same argument shows $J[h] = 0$ in two further cases. If $(\mathcal{L}_T + i\sigma)h = 0$ with $\sigma \neq 0$, then $\mathcal{L}_T J[h] = -i\sigma J[h]$, while $\mathcal{L}_T$ annihilates every Killing field of KdS. If $\mathcal{L}_T h = 0$ and h has vanishing φ-average, then $J[h]$ is a constant combination of $\partial_t, \partial_\varphi$, hence equal to its own φ-average, which is J of the φ-average of h, namely 0.

Consequently $\phi\rho = \bar{\phi}\bar{\rho}$, $\phi\rho' = \bar{\phi}\bar{\rho}'$ and $\phi\tau = -\bar{\phi}\bar{\tau}'$ hold to first order.

7.3. **Carter's Killing field.** Let $R = \operatorname{Re}\phi$ and $I = \operatorname{Im}\phi$. These are the Carter coordinates r and $-a\cos\theta$. Reality gives

$$\text{Þ}R = -A, \quad \text{Þ}'R = B, \quad \text{ð}R = 0, \quad \text{Þ}I = \text{Þ}'I = 0, \quad \text{ð}I = iC.$$

Define

$$\zeta = I^2(Bl + An) + R^2(Dm + C\bar{m}).$$

Lemma 7.6. *To first order,* ζ *is Killing. On KdS,* $\zeta = -\Xi(a^2\partial_t + a\partial_\varphi)$.

Proof. The ll, nn, mm and $\bar{m}\bar{m}$ components are immediate from those of ξ. The ln component equals $-|\phi|^2\phi(\tau\bar{\tau} - \tau'\bar{\tau}') = 0$, since by reality $\tau\bar{\tau} = \tau'\bar{\tau}'$.

The $m\bar{m}$ component equals $|\phi|^2\phi(\rho\bar{\rho}' - \rho'\bar{\rho}) = 0$, again by reality.

The lm component equals $2AC\phi + |\phi|^2(\text{ð}A + \bar{\tau}'A) = 0$, using $\text{ð}A = -\phi\bar{\rho}\tau$.

The nm component equals $-2BC\bar{\phi} + |\phi|^2(\text{ð}B + \tau B)$. Differentiating the first-order reality relation $\phi\rho' = \bar{\phi}\bar{\rho}'$ along m gives $\text{ð}B + \tau B = 2\tau B$, and the sum vanishes. The remaining components are complex conjugates. The value on KdS is an exact computation (item C6). □

7.4. **Pure gauge modes.** On KdS, $\partial_t = -\xi/\Xi$ and $\partial_\varphi = (a^2\xi - \zeta)/(\Xi a)$. Hence there are constants α_T, β_T and $\alpha_\varphi, \beta_\varphi$ with

$$T = \alpha_T\xi + \beta_T\zeta, \qquad \partial_\varphi = \alpha_\varphi\xi + \beta_\varphi\zeta.$$

Whenever $J[h] = 0$, Lemmas 7.3 and 7.6 give $\mathcal{L}_\xi h = -\mathcal{L}_{\delta\xi} g$ and $\mathcal{L}_\zeta h = -\mathcal{L}_{\delta\zeta} g$, and therefore

$$\mathcal{L}_T h = -\mathcal{L}_{Y_T} g, \qquad \mathcal{L}_{\partial_\varphi} h = -\mathcal{L}_{Y_\varphi} g, \qquad Y_\bullet = \alpha_\bullet\delta\xi + \beta_\bullet\delta\zeta. \tag{7.1}$$

The vector fields Y_T and Y_φ are local linear expressions in h. They are smooth across the horizons: ξ, ζ, $\delta\xi$ and $\delta\zeta$ are boost-spin invariant constructions from the perturbed Weyl spinor, so they can be computed in a horizon-regular principal tetrad, and $\Psi_2 \neq 0$ on $r > 0$.

Proposition 7.7. *(i) If* $(\mathcal{L}_T + i\sigma)h = 0$ *with* $\sigma \neq 0$, *then* $h = \mathcal{L}_X g$ *with* $X = Y_T/(i\sigma)$.

(ii) If $\mathcal{L}_T h = 0$ *and* h *has vanishing* φ*-average, then* $h = \mathcal{L}_X g$, *where* X *is the unique* φ*-antiderivative of* $-Y_\varphi$ *with vanishing* φ*-average.*

In both cases X *is smooth, and* $(\mathcal{L}_T + i\sigma)X = 0$.

Proof. In both cases $J[h] = 0$ by Remark 7.5, so (7.1) holds.

(i) Here $-i\sigma h = \mathcal{L}_T h = -\mathcal{L}_{Y_T} g$.

(ii) The field Y_φ has vanishing φ-average, because Y_φ commutes with rotations and h has vanishing average. So its mean-zero φ-antiderivative X (with $[\partial_\varphi, X] = -Y_\varphi$) is smooth. Then $\mathcal{L}_{\partial_\varphi}(h - \mathcal{L}_X g) = -\mathcal{L}_{Y_\varphi} g + \mathcal{L}_{Y_\varphi} g = 0$. So $h - \mathcal{L}_X g$ is axisymmetric with vanishing average, hence zero. □

7.5. **The stationary axisymmetric sector.**

Proposition 7.8. *If $\mathcal{L}_{\partial_t} h = \mathcal{L}_{\partial_\varphi} h = 0$ and $\delta\Psi_0 = \delta\Psi_4 = 0$, then $h = c_M \partial_M g + c_a \partial_a g + \mathcal{L}_X g$, with $X = X_0 + t_* K$, X_0 stationary and K a Killing field.*

Proof. Step 1: restoring reality. Here $J[h] = A'\partial_t + B'\partial_\varphi$ need not vanish. A constant phase change of ϕ removes A'. Subtracting a multiple of the tangent to the Plebański–Demiański family in the direction of the acceleration removes B': this is the only Plebański–Demiański tangent that makes ξ complex, and its J has nonzero ∂_φ-part (exact nonlinear check, item C6). After this, reality holds, and ξ and ζ are first-order Killing fields. They commute, since $\delta\xi$ and $\delta\zeta$ are background Killing fields.

Step 2: Carter's ansatz. One has $\nabla R \perp \nabla I$. Put $\Delta_R := -2AB|\phi|^2$ and $\Delta_I := -2CD|\phi|^2$. Then Δ_R depends on R alone and Δ_I on I alone, since $\eth(AB|\phi|^2) = 0$ by reality and $\xi(\Delta_R) = 0$. All the inner products of $\xi, \zeta, \nabla R$ and ∇I are functions of $(R, I, \Delta_R, \Delta_I)$. In coordinates adapted to (ξ, ζ, R, I), the perturbed metric is therefore, to first order, Carter's separable ansatz

$$\frac{S}{\Delta_R} dR^2 + \frac{S}{\Delta_I} dI^2 + \frac{\Delta_I (d\tilde{t} + R^2 d\psi)^2 - \Delta_R (d\tilde{t} - I^2 d\psi)^2}{S}, \qquad S = R^2 + I^2.$$

Step 3: the field equations. On this ansatz the Λ-vacuum equations are linear in (Δ_R, Δ_I), with general solution

$$\Delta_R = -\tfrac{\Lambda}{3} R^4 + \varepsilon R^2 - 2\mathfrak{m} R + \gamma, \qquad \Delta_I = -\tfrac{\Lambda}{3} I^4 - \varepsilon I^2 + 2\mathfrak{n} I + \gamma$$

(exact computation, item C7). Modulo the rescaling gauge, the variations are δM, δa and the NUT variation $\delta\mathfrak{n}$, together with the acceleration removed in Step 1.

Step 4: axis regularity. Write the background as $\Delta_I = (a^2 - I^2)(c + \tilde{\lambda} I^2)$ with $\tilde{\lambda} = \Lambda/3$ and $c > 0$, so the axes are the roots $I = \pm a$ and $\Delta_I'(\pm a) = \mp 2a(c + \tilde{\lambda} a^2)$.

(i) Misner string. At a simple root I_0 of Δ_I, the Killing field that vanishes on the axis is $\partial_\psi + I_0^2 \partial_{\tilde{t}}$, as one reads off from the ansatz. Under $\delta\Delta_I = 2\,\delta\mathfrak{n}\, I$ both roots move by the same amount: $\delta I_\pm = -2\delta\mathfrak{n}\, I_\pm / \Delta_I'(I_\pm) = \delta\mathfrak{n}/(c + \tilde{\lambda} a^2)$, since Δ_I' is odd. So the two axis generators differ by $4a\,\delta I_+\,\partial_{\tilde{t}}$. Because $\tilde{t}$ is not periodic on M, both axes are regular only if the generators coincide, which forces $\delta\mathfrak{n} = 0$ (as $a \neq 0$). The acceleration tangent does not move the axis generators at first order, since its $\delta\Delta_I$ is even in I after the conformal rescaling.

(ii) Conical regularity. With $\delta\mathfrak{n} = 0$, the ratio of the two conical periods along the acceleration tangent is $1 + 4M\,\delta\alpha_{\mathrm{acc}} + O(\delta^2)$. This is the first-order expansion of the C-metric condition $P(0) = P(\pi)$ in Griffiths–Podolský form, $P = 1 - 2\alpha M \cos\theta + O(a^2, \Lambda a^2)\cos^2\theta$, where the $\cos^2\theta$ term is even. A single period of φ regularizes both axes only if $\delta\alpha_{\mathrm{acc}} = 0$.

So both variations vanish. The gauge vector X generates the change from the background to the perturbed adapted coordinates, together with the rescalings of ξ and ζ.

Step 5: the gauge vector on all of M. Here $\partial_M g, \partial_a g$ are the parameter derivatives in the regular coordinates $(t_*, r, \theta, \varphi_*)$ [4]. By Lemma 7.1, Steps 1–4 apply on each of the three components of $M \setminus \{r = r_1, r_2\}$ (where $\Psi_2 \neq 0$, $\Delta_R \neq 0$), giving $h = c_M^{(i)} \partial_M g + c_a^{(i)} \partial_a g + \mathcal{L}_{X^{(i)}} g$. The constants agree: for $K \in \{\partial_t, \zeta\}$ the Iyer–Wald charge $\delta H_K[h]$ is the integral of a closed form smooth on M, hence the same on every sphere $\{t_* = \tau, r = \rho\}$; it vanishes on pure gauge terms

(proof of Theorem 1.2), and (E, J) has nonzero Jacobian in (M, a). So $k := h - c_M\partial_M g - c_a\partial_a g$ is smooth on M and equals $\mathcal{L}_{X^{(i)}}g$ on each region.

To glue, let S be the intersection of the axis with the horizons (codimension three). Away from the axis, write $X = X^r\partial_r + X^\theta\partial_\theta + Y$ with $Y = Y^t\partial_{t_*} + Y^\varphi\partial_{\varphi_*}$. Since $R = \operatorname{Re}\phi$ and $I = \operatorname{Im}\phi$ are natural, $X(R) = \delta R[k]$ and $X(I) = \delta I[k]$ are given by one local formula in k, which fixes X^r, X^θ. Then $\mathcal{L}_Y g = k - \mathcal{L}_{X^r\partial_r + X^\theta\partial_\theta}g = 2\,dY^t \odot \partial^\flat_{t_*} + 2\,dY^\varphi \odot \partial^\flat_{\varphi_*}$, and this determines (dY^t, dY^φ): if $\alpha \odot \partial^\flat_{t_*} + \beta \odot \partial^\flat_{\varphi_*} = 0$ with $\alpha(\partial_{\varphi_*}) = 0$, then $\alpha = y\,\partial^\flat_{\varphi_*}$ and $y\,g_{\varphi\varphi} = 0$. So off the axis the $X^{(i)}$ are restrictions of a single smooth field, up to Killing fields, which we absorb. Near the axis but off S, $X = X^{(i)}$ is smooth on each region. Hence X is smooth on $M \setminus S$ with $\mathcal{L}_X g = k$. Finally, $\mathcal{L}_X g = k$ implies the Killing transport system: $(X, \nabla X)$ satisfies, along any curve, a linear ODE with coefficients smooth in g, k and ∇k. Transporting from points off S extends X smoothly across S. So $k = \mathcal{L}_X g$ on M. The same applies to the t_*-linear parts in the generalized-mode induction below. □

7.6. **Generalized modes and proof of Theorem 1.2.**

Proof of Theorem 1.2. Part (1). Let $(\mathcal{L}_T + i\sigma)^k h = 0$ with $\sigma \neq 0$. We argue by induction on k; the case $k = 1$ is Proposition 7.7(i). For $k \geq 2$, the top coefficient $h_1 = (\mathcal{L}_T + i\sigma)^{k-1}h$ is a mode, so $h_1 = \mathcal{L}_{X_1}g$ with $(\mathcal{L}_T + i\sigma)X_1 = 0$. Put $Y = t_*^{k-1}X_1/(k-1)!$, which is smooth since t_* is regular. Then $(\mathcal{L}_T + i\sigma)^{k-1}Y = X_1$, so $h - \mathcal{L}_Y g$ has order $k - 1$.

Part (2). Let $(\mathcal{L}_T)^k h = 0$, and split $h = h_0 + h_\perp$ into its φ-average and the rest. The same induction, using Proposition 7.7(ii), shows that $h_\perp$ is pure gauge. The average h_0 satisfies $\mathcal{L}_{\partial_\varphi}h_0 = 0$ and $(\mathcal{L}_{\partial_t})^k h_0 = 0$. For $k = 1$, Proposition 7.8 applies. For $k \geq 2$: Let $h_1 := (\mathcal{L}_{\partial_t})^{k-1}h_0$. It is stationary and axisymmetric and solves the linearized equation, so by Proposition 7.8 $h_1 = c_M\partial_M g + c_a\partial_a g + \mathcal{L}_X g$. By Step 5 of that proof, $X = X_0 + t_*K$, where X_0 is stationary and $K = c^t\partial_{t_*} + c^\varphi\partial_{\varphi_*}$ is Killing.

The Iyer–Wald charges δH_K, for $K \in \{\partial_t, \zeta\}$, are independent of the 2-sphere. They vanish on $\mathcal{L}_Y g$ for every smooth Y: from $\Theta(\mathcal{L}_Y g) = Y \cdot L + dQ_Y$ and naturality of Q, one gets $\delta H_K[\mathcal{L}_Y g] = -\oint(Q_{[Y,K]} + Q_{[K,Y]}) = 0$. Since $h_1 = (\mathcal{L}_{\partial_t})^{k-1}h_0$ with $k \geq 2$, we have $\delta H_K[h_1] = \frac{d}{d\tau}\delta H_K[(\mathcal{L}_{\partial_t})^{k-2}h_0]|_{\{t_*=\tau\}} = 0$, because the charge of a solution does not depend on τ. Hence $\delta H_K[c_M\partial_M g + c_a\partial_a g] = 0$. Since $(E, J) = (M/\Xi^2, aM/\Xi^2)$ has Jacobian $M/\Xi^4 \neq 0$ with respect to (M, a), it follows that $c_M = c_a = 0$.

Put $Z = \frac{t_*^{k-1}}{(k-1)!}X_0 + \frac{t_*^k}{k!}K$, which is smooth since t_* is regular. Because $[\partial_{t_*}, X_0] = [\partial_{t_*}, K] = 0$ and $\partial_{t_*}t_* = 1$, we get $(\mathrm{ad}_{\partial_{t_*}})^{k-1}Z = X_0 + t_*K = X$. Hence $(\mathcal{L}_{\partial_t})^{k-1}(h_0 - \mathcal{L}_Z g) = h_1 - \mathcal{L}_X g = 0$, so $h_0 - \mathcal{L}_Z g$ has order $k - 1$, and the induction proceeds. □

Appendix A. Computer-assisted steps

Except for C9, all computations are exact over $\mathbb{Q}$ or $\mathbb{Q}(i)$. A Bernstein certificate on a box means that each relevant polynomial factor has Bernstein coefficients of one sign, not all zero.

C1. *Dual potential.* After Vieta elimination, the numerators of $n_1 - (\lambda C_0 + c_1)$ and $n_2 + 3$ are divisible by $s^2 - 4$.

C2. *Window margin.* With $\lambda = \lambda_{\mathrm{cr}}$ and a^2 reduced modulo $b^2 - e_2 b + e_4$, the sign conditions $F_1 > 0$, $P = F_1^2(e_2^2 - 4e_4) - Y^2 > 0$ for Q_2, $m = 1$ and $m = 2$ hold by a single-box Bernstein certificate on $[0, 1]^3$ in $(r_0, 1/r_2, \tau)$, for every irreducible factor (degrees up to $(34, 36, 8)$).

C3. *One sign change.* The combinations g_1, g_2 of Lemma 4.3, factored and certified in two variables.

C4. *Seam inequality.* In $(0, 1)^4$: $\partial_A N < 0$ (one box); $N \geq 0 \Rightarrow Q > 0$ at $A = 0$ and at $A = A_*$ (51 and 25 boxes), and at $\rho = 1$; on each box one of the two polynomials is certified.

C5. *Zero frequency.* The Teukolsky–Starobinsky identity at $K = 0$ and the 4×4 determinant.
C6. *Type D identities.* Exact checks, on KdS and on the accelerating NUT Plebański–Demiański metric, of identities derived by hand.
C7. *Carter ansatz.* The Ricci tensor of the ansatz with symbolic $\Delta_R(R), \Delta_I(I)$, and the solution of the resulting linear equations.
C8. *SdS Teukolsky–Starobinsky identities* (6.2): the first as an operator identity, the second by reduction modulo the radial equation (the R' coefficient vanishes and the R coefficient is the constant $\mathcal{C}_\ell$).
C9. *Cosmological side* (Proposition 3.10). In $(x, v, \tau) = (r_0, 1/r_2, \tau) \in (0,1)^3$, $b = a^2$ enters through the root b_+ of $b^2 - e_2 b + e_4$ (a smooth cubic curve, so no rational parametrisation). Signs at b_+ come from the reduction $F_1 b + F_0$ via (F_1, $Y = -(2F_0 + F_1 e_2)$, $P = F_1^2 D - Y^2$), or, for large r_2, from interval evaluation of the unreduced polynomial on an interval enclosure of $b_+ = 2e_4 v^2/(e_2 v^2 - \sqrt{D v^4})$. The atlas is the cube with projective blow-up charts at the faces $x = 1$, $v = 0$, $v = 1$, $\tau = 1$ and their intersections. Conditions: (i) $L(t)$ at $m = 1, 2$ and its m^2-coefficient are positive where $2\delta \geq t - 1$ (exact); (ii) $2q_1^2\theta - 1.05\, t(t-1)(-L(t)) \geq 0$ for $m = 1, 2$ and the m^2-coefficient, exactly outside $\{x, v \geq 7/8\}$; inside, (i) and $A \geq 0$ reduce it to a quadratic inequality in τ; (iii), (iv) with the alternative "$\varpi_m \leq 0$" on each box; (v) on a single box. *Rounding.* Outside (i) and the exact part of (ii), coefficients are computed exactly, normalized, rounded once to double precision, and subdivided by de Casteljau midpoints. Carrying alongside each array A the array B obtained by the same subdivisions of $|A_{\mathrm{root}}|$, each entry satisfies $|A_{\mathrm{float}} - A_{\mathrm{exact}}| \leq (1+u)^{1+\Sigma} u(1+\Sigma)B \leq 1.01\, u(1+\Sigma)B$ (as $u\Sigma < 10^{-10}$), where $u = 2^{-53}$ and Σ is the sum of the degrees subdivided. A sign is accepted only if $\pm A > 10^{-13}(k+4)B$ entrywise at depth k; since all degrees are at most 122, this exceeds the error bound by a factor of at least 7.
C10. *Threshold Teukolsky–Starobinsky map* (Lemma 3.3). With $\Delta_r R'' + (s+1)\Delta_r' R' + Q_s R = 0$, Q_s as in [11], the function $\Delta_r^2 (D^\dagger)^4 (\Delta_r^2 R)$ solves the $s = -2$ equation, symbolically in $(\Lambda, M, a, \sigma, m, \lambda)$, after reduction modulo the $s = +2$ equation.

Statement on the use of AI

In preparing this work I used Claude Opus 5.5 (Anthropic) to develop and check proofs, write and run the symbolic and certificate code, search and check the literature, and draft and edit the text. The proofs were also checked in several adversarial review passes, each run by an AI model. These passes found errors, including invalid earlier versions of the large-A regime of the event lemma and of the passage to the upper half plane, and every error found was corrected. Numerical experiments were used only to guide the search for proofs, and no result depends on them. No human expert has yet checked the argument. I have reviewed and edited the content and take full responsibility for it.

Data availability

The code for the computer-assisted steps is available from the author on request.

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